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Data Structures

Dsu

```

/* DSU (Disjoint Set Union / Union-Find)
 * Mantem conjuntos disjuntos com uniao e busca eficiente.
 * Complexidade: O(a(N)) por operacao, onde a e a inversa de Ackermann.
 * Memoria: O(N)
 * Metodos:
 * - find(i): Retorna o representante do conjunto de i.
 * - join(i, j): Une os conjuntos de i e j, retorna true se uniu.
 * - same(i, j): Verifica se i e j estao no mesmo conjunto.
 * - size(i): Retorna o tamanho do conjunto de i.
 */
D56 struct DSU {
1A8     int n;
923     vector<int> parent;
1DF     vector<int> sz;
75E     int num_sets;

DDB     DSU(int _n) : n(_n), parent(_n), sz(_n, 1), num_sets(_n) {
4D6         iota(parent.begin(), parent.end(), 0);
548     }

C42     int find(int i) {
331         if (parent[i] == i) return i;
565         return parent[i] = find(parent[i]);
72C     }

D58     bool join(int i, int j) {
477         i = find(i), j = find(j);
41E         if (i != j) {
4C2             if (sz[i] < sz[j]) swap(i, j);
1F0             parent[j] = i;
529             sz[i] += sz[j];
CA8             num_sets--;
8A6             return true;
20F         }
D1F         return false;
30C     }

00C     int size(int i) { return sz[find(i)]; }
DA3     bool same(int i, int j) { return find(i) == find(j); }
FF6 };

```

Mo

```

/* Mo's Algorithm - O((N + Q) * sqrt(N))
 * queries: vetor de {l, r} 0-indexado, fechado.
 * add(i), rem(i): adicionam/removem o elemento i da janela.
 * ans(qi): registra resposta para a query de indice qi.
 *
 * Exemplo:
 * vector<int> ans(q);
 * Mo mo(n, queries);
 * mo.run(
 *     [&](int i) { ... },

```

```

*     [&](int i) { ... },
*     [&](int qi) { ans[qi] = ...; }
* );
*/
0DF struct Mo {
7F4     int n, block;
304     struct Query { int l, r, idx; };
240     vector<Query> queries;

8CA     Mo(int n, vector<pair<int,int>>& qs)
10A         : n(n), block(max(1, (int)sqrt(n))) {
D26         for (int i = 0; i < (int)qs.size(); i++)
9AA             queries.push_back({qs[i].first, qs[i].second, i});
F94         sort(queries.begin(), queries.end(),
724             [&](const Query& a, const Query& b) {
ED6                 int ba = a.l / block, bb = b.l / block;
A2D                 if (ba != bb) return ba < bb;
FDF                 return (ba & 1) ? a.r > b.r : a.r < b.r;
FC3             });
8F7     }

D5A     template<typename Add, typename Rem, typename Ans>
743     void run(Add add, Rem rem, Ans ans) {
70A         int L = 0, R = -1;
FFE         for (auto& [l, r, idx] : queries) {
DBE             while (R < r) add(++R);
EBF             while (L > l) add(--L);
153             while (R > r) rem(R--);
319             while (L < l) rem(L++);
414             ans[idx];
4B3         }
1D9     }
09E };

/* HilbertMo - Mo com ordenacao pela curva de Hilbert
 * Mesma interface que Mo, mas com constante menor na pratica.
 */
CD0 struct HilbertMo {
22A     struct Query { int l, r, idx; long long ord; };
240     vector<Query> queries;

B0A     static long long hilbert(int x, int y, int n) {
CEE         long long d = 0;
F8A         for (int s = n >> 1; s >= 1) {
7B9             int rx = (x & s) > 0, ry = (y & s) > 0;
065             d += (long long)s * s * ((3 * rx) ^ ry);
069             if (!ry) {
1FD                 if (rx) { x = s - 1 - x; y = s - 1 - y; }
9DD                 swap(x, y);
FFF             }
070         }
BE2         return d;
151     }

E76     HilbertMo(int n, vector<pair<int,int>>& qs) {
ABE         int hn = 1;
FCC         while (hn < n) hn <= 1;
D26         for (int i = 0; i < (int)qs.size(); i++)
F80             queries.push_back({qs[i].first, qs[i].second, i,
C04                 hilbert(qs[i].first, qs[i].second, hn)});
F94         sort(queries.begin(), queries.end(),
B18             [&](const Query& a, const Query& b) {
73A                 return a.ord < b.ord;
A3A             });
0D6     }

D5A     template<typename Add, typename Rem, typename Ans>
743     void run(Add add, Rem rem, Ans ans) {
70A         int L = 0, R = -1;
277         for (auto& [l, r, idx, ord] : queries) {
DBE             while (R < r) add(++R);
EBF             while (L > l) add(--L);
153             while (R > r) rem(R--);
319             while (L < l) rem(L++);
414             ans[idx];
F5F         }
EBB     }
EF4 };

```

Fenwick Tree

```

/* Fenwick Tree (Point Update, Prefix Query)
 * Realiza atualizacoes pontuais e consultas de prefixo.
 * Complexidade: O(log N) para update e query.
 * Memoria: O(N)
 * Requisitos:
 * - NODE deve ter: operator+=, operator- (para range query) e
 *   construtor default.
 */
/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}

```

```

void operator+=(const Node& other) { val += other.val; }
Node operator-(const Node& other) const { return Node(val - other.val); }
};
*/

8A0 template<typename NODE>
0F1 struct FenwickTree {
060     int N;
9B8     vector<NODE> bit;
5D2     FenwickTree(int n) : N(n), bit(n + 1) {}
D6B     void update(int idx, NODE v) {
B2F         for (idx++; idx <= N; idx += idx & -idx) {
046             bit[idx] += v;
23E         }
9DA     }
FCA     NODE query(int idx) { // [0, idx]
ADA         NODE res;
159         for (idx++; idx > 0; idx -= idx & -idx) {
8E6             res += bit[idx];
A38         }
B50         return res;
C00     }
762     NODE query(int l, int r) { // [l, r] (precisa ser reversivel)
7C3         if (l > r) return NODE();
A4E         return query(r) - query(l - 1);
821     }
134 };

```

Fenwick Tree 2D

```

/* Fenwick Tree 2D (Point Update, Rectangle Query)
* Realiza atualizacoes pontuais e consultas de prefixo em matrizes.
* Complexidade: O(log N * log M) para update e query.
* Memoria: O(N * M)
* Requisitos:
* - NODE deve ter: operator+==, operator- e construtor default.
*/

/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) { val += other.val; }
    Node operator-(const Node& other) const { return Node(val - other.val); }
};
*/

8A0 template<typename NODE>
2E6 struct FenwickTree2D {
610     int N, M;
803     vector<vector<NODE>>> bit;
237     FenwickTree2D(int n, int m) : N(n), M(m), bit(n + 1, vector<NODE>(m + 1)) {}
274     void update(int r, int c, NODE v) { // point update
3BA         for (int i = r + 1; i <= N; i += i & -i) {
786             for (int j = c + 1; j <= M; j += j & -j) {
9ED                 bit[i][j] += v;
95F             }
A1F         }
BF9     }
82E     NODE query(int r, int c) { // [(0,0), (r,c)]
ADA         NODE res;
8D9         for (int i = r + 1; i > 0; i -= i & -i) {
83F             for (int j = c + 1; j > 0; j -= j & -j) {
01D                 res += bit[i][j];
CB5             }
E20         }
B50         return res;
943     }
05F     NODE query(int r1, int c1, int r2, int c2) { // [(r1,c1), (r2,c2)]
5C4         if (r1 > r2 || c1 > c2) return NODE();
B7F         return query(r2, c2) - query(r1 - 1, c2)
794             - query(r2, c1 - 1) + query(r1 - 1, c1 - 1);
OE7     }
FF7 };

```

Dynamic Fenwick Tree 2D

```

/* Fenwick Tree 2D Sparse (Point Update, Rectangle Query)
* Suporta coordenadas ate 10^9 usando compressao interna.
* Complexidade: Build O(P log P), Update/Query O(log^2 P).
* Memoria: O(P log P).
* Requisitos:
* - NODE deve ter: operator+==, operator-, operator+ e construtor default.
* - Offline: Todos os pontos de interesse devem ser passados no build.
*
* Creditos: Adaptado de TFG (Thiago Ferreira Goncalves)
* Fonte: github.com/tfg50/Competitive-Programming
*/

/* --- Exemplo de NODE (Soma com Inclusao-Exclusao) ---

```

```

struct Node {
    ll val;
    Node(long long v = 0) : val(v) {}
    void operator+=(const Node& other) {
        val += other.val;
    }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
    Node operator+(const Node& other) const {
        return Node(val + other.val);
    }
};
*/

8A0 template<typename NODE>
222 struct FenwickTree2DSparse {
6C1     vector<int> ord;
803     vector<vector<NODE>>> bit;
E76     vector<vector<int>>> coord;
BEF     FenwickTree2DSparse(vector<pii> pts) {
CD7         sort(pts.begin(), pts.end());
5A8         for (auto [x, y] : pts) {
DC4             if (ord.empty() || x != ord.back()) ord.push_back(x);
ABA         }
261         bit.resize(ord.size() + 1);
3EB         coord.resize(ord.size() + 1);
A0A         sort(pts.begin(), pts.end(), [](const pii& a, const pii& b) {
463             return a.second < b.second;
19C         });
5A8         for (auto [x, y] : pts) {
A1A             for (int i = get_x(x); i < (int)bit.size(); i += i & -i) {
3E1                 if (coord[i].empty() || coord[i].back() != y)
739                     coord[i].push_back(y);
D32             }
BAF         }
BE8         for (int i = 0; i < (int)bit.size(); i++) {
F3E             bit[i].assign(coord[i].size() + 1, NODE());
B33         }
84E     }
F1C     inline int get_x(int x) {
2C8         return upper_bound(ord.begin(), ord.end(), x) - ord.begin();
FF7     }
DD7     inline int get_y(int i, int y) {
190         return upper_bound(coord[i].begin(), coord[i].end(), y)
7A2             - coord[i].begin();
B44     }
5CF     void update(int x, int y, NODE v) {
A1A         for (int i = get_x(x); i < (int)bit.size(); i += i & -i) {
511             for (int j = get_y(i, y); j < (int)bit[i].size(); j += j & -j) {
9ED                 bit[i][j] += v;
66B             }
638         }
0CE     }
84C     NODE query(int x, int y) {
ADA         NODE res;
FAD         for (int i = get_x(x); i > 0; i -= i & -i) {
263             for (int j = get_y(i, y); j > 0; j -= j & -j) {
01D                 res += bit[i][j];
359             }
92F         }
B50         return res;
7C4     }
FCA     NODE query(int x1, int y1, int x2, int y2) {
CD6         return query(x2, y2) - query(x1 - 1, y2)
CAD             - query(x2, y1 - 1) + query(x1 - 1, y1 - 1);
944     }
67D };

```

Sparse Table

```

/* Sparse Table (Static RMQ)
* Realiza consultas em range em tempo constante.
* Complexidade: O(N log N) no build, O(1) na query.
* Memoria: O(N log N)
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e construtor.
* - A operacao DEVE ser idempotente: f(a, a) = a.
*/

/* --- Exemplo de NODE (Indice do Minimo) ---
struct Node {
    int val, pos;
    Node(int v = 2e9, int p = -1) : val(v), pos(p) {}
    static inline Node merge(const Node& l, const Node& r) {
        return l.val <= r.val ? l : r;
    }
};

```

```

};

// Uso: SparseTable<NODE> st(v) onde v e vector<NODE>
// for (int i = 0; i < n; i++) { cin >> x; v[i] = Node(x, i); }
*/

8A0 template<typename NODE>
7E9 struct SparseTable {
5F0     int N, K;
6F5     vector<vector<NODE>>> st;
4A7     vector<int> lg;
67A     template<typename T>
E4A     SparseTable(const vector<T& v) : N(v.size()) {
A77         K = (N > 0) ? __lg(N) : 0;
3A9         st.assign(K + 1, vector<NODE>(N));
641         lg.assign(N + 1, 0);
BB5         for (int i = 2; i <= N; i++) lg[i] = lg[i / 2] + 1;
DDD         for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
2CE         for (int j = 1; j <= K; j++)
285             for (int i = 0; i + (1 << j) <= N; i++)
4E1                 st[j][i] = NODE::merge(st[j - 1][i],
A64                                         st[j - 1][i + (1 << (j - 1))]);
CE2     }
762     NODE query(int l, int r) {
2C3         int j = lg[r - l + 1];
8F2         return NODE::merge(st[j][l], st[j][r - (1 << j) + 1]);
843     }
231 };

```

Disjoint Sparse Table

```

/* Disjoint Sparse Table (Static Range Query)
* Realiza consultas em range em O(1) para qualquer operacao associativa.
* Complexidade: O(N log N) no build, O(1) na query.
* Memoria: O(N log N)
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e construtor.
* - Operacao deve ser ASSOCIATIVA: f(f(a,b),c) = f(a,f(b,c)).
*/

/* --- Exemplo de NODE (Soma em Range) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    static inline Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};
*/

8A0 template<typename NODE>
906 struct DisjointSparseTable {
5F0     int N, K;
6F5     vector<vector<NODE>>> st;
67A     template<typename T>
68E     DisjointSparseTable(const vector<T& v) : N(v.size()) {
3AA         if (N == 0) return;
475         K = __lg(2 * N - 1);
C78         st.assign(K, vector<NODE>(1 << K));
DDD         for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
C54         for (int j = 1; j < K; j++) {
1C0             int len = 1 << j;
C2C             for (int m = len; m <= (1 << K); m += 2 * len) {
A34                 if (m < N) st[j][m] = st[0][m];
6D6                 for (int i = m + 1; i < min(N, m + len); i++)
0EA                     st[j][i] = NODE::merge(st[j][i - 1], st[0][i]);
C2E                 if (m - 1 < N) st[j][m - 1] = st[0][m - 1];
226                 for (int i = m - 2; i >= m - len; i--)
604                     st[j][i] = NODE::merge(st[0][i], st[j][i + 1]);
643             }
3F0         }
4D7     }
1F9     inline NODE query(int l, int r) {
434         if (l == r) return st[0][l];
B89         int j = __lg(l ^ r);
894         return NODE::merge(st[j][l], st[j][r]);
DB5     }
4D6 };

```

Segment Tree

```

/* Segment Tree (Point Update, Range Query)
* Complexidade: O(log N) para update e query.
* Memoria: O(4*N)
* Requisitos:
* - NODE deve ter: static merge(L, R), apply(V), e construtor identidade.
*/

/* --- Exemplo de NODE (Soma com Update de Substituicao) ---
struct Node {
    ll val = 0;
};

```

```

Node(ll v = 0) : val(v) {}
static inline Node merge(const Node& l, const Node& r) {
    return Node(l.val + r.val);
}
// Para Point Set: val = v;
// Para Point Add: val += v;
inline void apply(int v) {
    val = v;
}
};
*/

8A0 template<typename NODE>
383 struct SegTree {
060     int N;
B1C     vector<NODE> seg;
E7A     SegTree(int n) : N(n), seg(4 * n) {}
67A     template<typename T>
373     SegTree(const vector<T>& v) : N(v.size()), seg(4 * v.size()) {
231         build(1, 0, N - 1, v);
EF1     }
67A     template<typename T>
5B9     void build(int no, int l, int r, const vector<T>& v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }
663     void update(int no, int l, int r, int idx, int val) {
893         if (l == r) {
D88             seg[no].apply(val);
505             return;
EC9         }
27E         int m = (l + r) >> 1;
B73         if (idx <= m) update(no << 1, l, m, idx, val);
99C         else update((no << 1) | 1, m + 1, r, idx, val);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
358     }
38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),
074             query((no << 1) | 1, m + 1, r, a, b));
365     }
9B0     void update(int idx, int val) { update(1, 0, N - 1, idx, val); }
36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
45A };

```

Lazy Segment Tree

```

/* Lazy Segment Tree (Range Update, Range Query)
 * Realiza atualizacoes e consultas em range.
 * Complexidade: O(log N) para update e query.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R) e
 *   construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 */

/* --- Exemplo de NODE e TAG (Soma e Multiplicacao com Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void inline compose(const Tag& t) {
        add = (add * t.mul + t.add) % MOD;
        mul = (mul * t.mul) % MOD;
    }
};

struct Node {
    ll val = 0;
    Node(ll v = 0) : val(v) {}
    static inline Node merge(const Node& l, const Node& r) {
        return Node((l.val + r.val) % MOD);
    }
    void inline apply(const Tag& t, int l, int r) {
        val = (val * t.mul + t.add * (r - l + 1)) % MOD;
    }
};
*/

708 template<typename NODE, typename TAG>
2BA struct LazySegmentTree {
060     int N;
B1C     vector<NODE> seg;

```

```

71A     vector<TAG> lazy;
F0F     LazySegmentTree(int n) : N(n), seg(4 * n), lazy(4 * n) {}
67A     template<typename T>
373     LazySegmentTree(const vector<T>& v)
9EB         : N(v.size()), seg(4 * v.size()), lazy(4 * v.size()) {
231         build(1, 0, N - 1, v);
5EE     }
67A     template<typename T>
5B9     void build(int no, int l, int r, const vector<T>& v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }
F08     void push(int no, int l, int r) {
27E         int m = (l + r) >> 1;
1EA         int e = no << 1, d = e | 1;
25F         seg[e].apply(lazy[no], l, m);
115         lazy[e].compose(lazy[no]);
4E1         seg[d].apply(lazy[no], m + 1, r);
7CC         lazy[d].compose(lazy[no]);
47F         lazy[no] = TAG();
6A3     }
130     void update(int no, int l, int r, int a, int b, const TAG& v) {
2E6         if (b < l || r < a) return;
02B         if (a <= l && r <= b) {
8C7             seg[no].apply(v, l, r);
92C             lazy[no].compose(v);
505             return;
81F         }
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
9AB         update(no << 1, l, m, a, b, v);
087         update((no << 1) | 1, m + 1, r, a, b, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
A1A     }
38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),
074             query((no << 1) | 1, m + 1, r, a, b));
80A     }
0D7     void update(int l, int r, const TAG& v) { update(1, 0, N - 1, l, r, v); }
36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
54E };

```

Dual Segment Tree

```

/* Dual Segment Tree (Range Update, Point Query)
 * Realiza atualizacoes em range e consultas pontuais.
 * Complexidade: O(log N) para update e get.
 * Memoria: O(4*N)
 * Requisitos:
 * - TAG deve ter: compose(TAG), apply(T&) e construtor identidade.
 * - T e o tipo do dado nas folhas.
 * Exemplo de uso: DualSegTree<ll, MyTag> st(v);
*/

/* --- Exemplo de TAG (Soma e Multiplicacao com Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void compose(const Tag& t) {
        mul = (mul * t.mul) % MOD;
        add = (add * t.mul + t.add) % MOD;
    }
    void apply(ll& val) { val = (val * mul + add) % MOD; }
};
*/

2E5 template<typename T, typename TAG>
9C1 struct DualSegTree {
060     int N;
71A     vector<TAG> lazy;
E4B     vector<T> leaves;
11D     DualSegTree(int n) : N(n), lazy(4 * n), leaves(n) {}
5F6     DualSegTree(const vector<T>& v)
62C         : N(v.size()), lazy(4 * v.size()), leaves(v) {}
F08     void push(int no, int l, int r) {
F7F         int e = no << 1;
EE1         int d = e | 1;
115         lazy[e].compose(lazy[no]);

```

```

7CC         lazy[d].compose(lazy[no]);
47F         lazy[no] = TAG();
48C     }
130     void update(int no, int l, int r, int a, int b, const TAG& v) {
2E6         if (b < l || r < a) return;
02B         if (a <= l && r <= b) {
92C             lazy[no].compose(v);
505             return;
CA0         }
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
9AB         update(no << 1, l, m, a, b, v);
087         update((no << 1) | 1, m + 1, r, a, b, v);
667     }
629     T get(int no, int l, int r, int idx) {
893         if (l == r) {
4C5             T res = leaves[idx];
2E3             lazy[no].apply(res);
B50             return res;
BE7         }
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
52E         if (idx <= m) return get(no << 1, l, m, idx);
483         else return get((no << 1) | 1, m + 1, r, idx);
A24     }
A21     void update(int l, int r, const TAG& t) { update(1, 0, N - 1, l, r, t); }
AE4     T get(int idx) { return get(1, 0, N - 1, idx); }
6C2 };

```

Segment Tree Beats

```

/* Segment Tree Beats (Range Update, Range Query)
 * Extensao da Lazy Seg Tree com condicoes de parada e de tag.
 * Complexidade amortizada: O(N log^2 N) para sequencias de updates.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R),
 *   break_condition(TAG) e tag_condition(TAG),
 *   e construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 *
 * break_condition(tag): retorna true se o update pode ser ignorado
 *   (o no ja satisfaz a condicao - ex: mx <= v).
 * tag_condition(tag): retorna true se a tag pode ser propagada
 *   diretamente sem descer (ex: mx2 < v <= mx).
 */

/* --- Exemplo de NODE e TAG (chmin em range + query de soma/max) ---
struct Tag {
    ll val = LLONG_MAX; // identidade: nao faz nada
    void inline compose(const Tag& t) {
        val = min(val, t.val);
    }
};

struct Node {
    ll sum, mx, mx2;
    int cnt_mx, sz;

    Node() : sum(0), mx(LLONG_MIN), mx2(LLONG_MIN), cnt_mx(0), sz(0) {}
    Node(ll v) : sum(v), mx(v), mx2(LLONG_MIN), cnt_mx(1), sz(1) {}

    static inline Node merge(const Node& l, const Node& r) {
        Node res;
        res.sz = l.sz + r.sz;
        res.sum = l.sum + r.sum;
        if (l.mx == r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx + r.cnt_mx;
            res.mx2 = max(l.mx2, r.mx2);
        } else if (l.mx > r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx;
            res.mx2 = max(l.mx2, r.mx);
        } else {
            res.mx = r.mx; res.cnt_mx = r.cnt_mx;
            res.mx2 = max(l.mx, r.mx2);
        }
        return res;
    }

    // aplica a tag quando tag_condition e verdadeiro
    void inline apply(const Tag& t, int l, int r) {
        sum -= (ll)(mx - t.val) * cnt_mx;
        mx = t.val;
    }

    // ignora o update completamente (no ja satisfaz)
    bool inline break_condition(const Tag& t) const {
        return mx <= t.val;
    }

    // pode aplicar a tag diretamente sem descer
    bool inline tag_condition(const Tag& t) const {
        return mx2 < t.val;
    }
};

```

```

};
*/
708 template<typename NODE, typename TAG>
8C8 struct SegTreeBeats {
060     int N;
81C     vector<NODE> seg;
71A     vector<TAG> lazy;
8EA     SegTreeBeats(int n) : N(n), seg(4 * n), lazy(4 * n) {}
67A     template<typename T>
47B     SegTreeBeats(const vector<T& v)
9EB         : N(v.size()), seg(4 * v.size()), lazy(4 * v.size()) {
231         build(1, 0, N - 1, v);
5EE     }
67A     template<typename T>
5B9     void build(int no, int l, int r, const vector<T& v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
82E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }
F08     void push(int no, int l, int r) {
27E         int m = (l + r) >> 1;
1EA         int e = no << 1, d = e | 1;
25F         seg[e].apply(lazy[no], l, m);
115         lazy[e].compose(lazy[no]);
4E1         seg[d].apply(lazy[no], m + 1, r);
7CC         lazy[d].compose(lazy[no]);
47F         lazy[no] = TAG();
6A3     }
130     void update(int no, int l, int r, int a, int b, const TAG& v) {
FBB         if (b < l || r < a || seg[no].break_condition(v)) return;
40D         if (a <= l && r <= b && seg[no].tag_condition(v)) {
8C7             seg[no].apply(v, l, r);
92C             lazy[no].compose(v);
505             return;
2B2         }
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
9AB         update(no << 1, l, m, a, b, v);
087         update((no << 1) | 1, m + 1, r, a, b, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
18B     }
38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),
074             query((no << 1) | 1, m + 1, r, a, b));
80A     }
0D7     void update(int l, int r, const TAG& v) { update(1, 0, N - 1, l, r, v); }
36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
6F1 };

```

Dynamic Segment Tree

```

/* Dynamic Segment Tree with Lazy Propagation (Range Update, Range Query)
* Utilizada para intervalos gigantes (ex: 0 a 10^18).
* Complexidade: O(log N) por operacao.
* Memoria: O(Q log N) onde Q e o numero de operacoes.
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&),
*   apply(const TAG&, ll, ll) e construtor identidade.
* - TAG deve ter: apply(const TAG&), campo 'has' (bool) e
*   construtor identidade.
*/
/* --- Exemplo de NODE e TAG ---
struct Tag {
    ll add = 0;
    bool has = false;
    void apply(const Tag& t) {
        add += t.add;
        has = true;
    }
};

struct Node {
    ll val = 0;
    static Node merge(const Node& l, const Node& r) {
        return {l.val + r.val};
    }
    void apply(const Tag& t, ll l, ll r) {
        val += t.add * (r - l + 1);
    }
};

```

```

};
*/
708 template<typename NODE, typename TAG>
77D struct DynamicSegTree {
126     struct InternalNode {
4EB         NODE node;
657         TAG tag;
74F         int l, r;
FB6         InternalNode() : node(NODE()), tag(TAG()), l(-1), r(-1) {}
A61         void apply(const TAG& t, ll l, ll r) {
BE6             node.apply(t, l, r);
D21             tag.apply(t);
9C3         }
2FB     };
983     ll L, R;
vector<InternalNode> seg;
5C7     DynamicSegTree(ll l, ll r) : L(l), R(r) {
3BD         seg.reserve(4e6); seg.emplace_back();
BFD     }
4F9     void push(int v, ll l, ll r) {
A16         if (!seg[v].tag.has) return;
56B         if (l == r) return seg[v].tag = TAG();
579         if (seg[v].l == -1) { seg[v].l = seg.size(); seg.emplace_back(); }
203         if (seg[v].r == -1) { seg[v].r = seg.size(); seg.emplace_back(); }
769         ll mid = l + (r - l) / 2;
F3E         seg[seg[v].l].apply(seg[v].tag, l, mid);
0B7         seg[seg[v].r].apply(seg[v].tag, mid+1, r);
DD0         seg[v].tag = TAG();
AD7     }
352     void update(int v, ll l, ll r, ll ql, ll qr, const TAG& t) {
151         if (ql <= l && r <= qr) return seg[v].apply(t, l, r);
A48         push(v, l, r);
769         ll mid = l + (r - l) / 2;
441         if (ql <= mid) {
579             if (seg[v].l == -1) { seg[v].l = seg.size(); seg.emplace_back(); }
2DD             update(seg[v].l, l, mid, ql, qr, t);
0B1         }
64D         if (qr > mid) {
203             if (seg[v].r == -1) { seg[v].r = seg.size(); seg.emplace_back(); }
76B             update(seg[v].r, mid + 1, r, ql, qr, t);
216         }
C5A         NODE left = (seg[v].l != -1) ? seg[seg[v].l].node : NODE();
C56         NODE right = (seg[v].r != -1) ? seg[seg[v].r].node : NODE();
799         seg[v].node = NODE::merge(left, right);
15E     }
EE0     NODE query(int v, ll l, ll r, ll ql, ll qr) {
5CF         if (v == -1 || ql > r || qr < l) return NODE();
3C9         if (ql <= l && r <= qr) return seg[v].node;
A48         push(v, l, r);
769         ll mid = l + (r - l) / 2;
211         return NODE::merge(query(seg[v].l, l, mid, ql, qr),
0CC             query(seg[v].r, mid + 1, r, ql, qr));
933     }
4CA     void update(ll ql, ll qr, const TAG& t) { update(0, L, R, ql, qr, t); }
323     NODE query(ll ql, ll qr) { return query(0, L, R, ql, qr); }
4CA };

```

Dynamic Persistent Segment Tree

```

/* Persistent Dynamic Segment Tree (Point Update, Range Query)
* Permite consultas em versoes anteriores da arvore (persistencia).
* Aloca nos sob demanda, ideal para intervalos grandes e multiplas versoes.
* Complexidade: O(log N) por update e query.
* Memoria: O(Q log N) onde Q e o numero de updates.
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e
*   construtor identidade.
*/
/* --- Exemplo de NODE (Soma) ---
struct Node {
    int val;
    Node(int v = 0) : val(v) {}
    static inline Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};

8A0 template<typename NODE>
68A struct PersistentSegmentTree {
126     struct InternalNode {
1DF         NODE data;
74F         int l, r;
E61         InternalNode() : data(NODE()), l(0), r(0) {}
B12     };

```

```

060     int N;
788     vector<InternalNode> st;
34F     vector<int> roots;
BC9     PersistentSegmentTree(int n) : N(n) {
967         st.reserve(4e6); st.emplace_back();
405         roots.push_back(0);
71E     }
C93     int update(int prev_root, int L, int R, int idx, NODE v) {
4C0         int node = st.size();
6F5         if (prev_root == 0) st.emplace_back();
FFC         else st.push_back({st[prev_root]});
651         if (L == R) {
DC5             st[node].data = NODE::merge(st[node].data, v);
192             return node;
BE8         }
08E         int mid = L + (R - L) / 2;
D33         if (idx <= mid) { // Cuidado com ordem de avaliacao (LHS) pre-C++17
9DD             int prev_left = (prev_root == 0) ? 0 : st[prev_root].l;
5A1             st[node].l = update(prev_left, L, mid, idx, v);
0CC         }
4E6         else { // Cuidado com ordem de avaliacao (LHS) pre-C++17
709             int prev_right = (prev_root == 0) ? 0 : st[prev_root].r;
22D             st[node].r = update(prev_right, mid + 1, R, idx, v);
C59         }
777         st[node].data = NODE::merge(st[st[node].l].data, st[st[node].r].data);
192         return node;
F04     }
082     NODE query(int node, int L, int R, int i, int j) {
0B8         if (node == 0 || i > R || j < L) return NODE();
692         if (i <= L && R <= j) return st[node].data;
08E         int mid = L + (R - L) / 2;
B1E         return NODE::merge(query(st[node].l, L, mid, i, j),
521             query(st[node].r, mid + 1, R, i, j));
26E     }
D6B     void update(int idx, NODE v) {
69E         roots.push_back(update(roots.back(), 0, N - 1, idx, v));
1BF     }
3A5     NODE query(int version, int l, int r) {
191         return query(roots[version], 0, N - 1, l, r);
247     }
B19 };

```

Graphs

Dijkstra

```

/* Dijkstra – Caminho minimo de fonte unica
* Complexidade: O((V + E) log V)
* Requer pesos nao-negativos.
*
* adj[u] = lista de {v, w}: aresta u-v com peso w
*
* Retorna vetor dist onde dist[v] = distancia minima de s a v.
* dist[v] = INF se v e inalcançavel.
*/

```

```

67A template<typename T>
7D8 vector<T> dijkstra(const vector<vector<pair<int,T>>& adj, int s) {
B4C     int n = adj.size();
305     const T INF = numeric_limits<T>::max();
835     vector<T> dist(n, INF);
CFC     priority_queue<pair<T,int>, vector<pair<T,int>>, greater<>> pq;
A93     dist[s] = 0;
7BA     pq.push({0, s});
502     while (!pq.empty()) {
5C1         auto [d, u] = pq.top(); pq.pop();
3E1         if (d > dist[u]) continue;
610         for (auto [v, w] : adj[u])
DDE             if (dist[u] + w < dist[v]) {
491                 dist[v] = dist[u] + w;
BF3                 pq.push({dist[v], v});
D74             }
9AE         }
8D7         return dist;
25B     }

```

Euler Path

```

/* Caminho/Circuito Euleriano – Hierholzer – O(V + E)
*
* Condiçoes de existencia:
* - Grafo nao-direcionado:
*   - Circuito: todos os vertices com grau par.
*   - Caminho: exatamente 2 vertices com grau impar (inicio e fim).
* - Grafo direcionado:

```



```

* - Circuito: in_degree[v] == out_degree[v] para todo v.
* - Caminho: exatamente 1 vertice com out-in=+1 (inicio),
*             exatamente 1 com in-out=+1 (fim), resto iguais.
*
* euler(adj, s) -> vetor de vertices do caminho (tamanho E+1).
* Retorna vetor vazio se nao existe caminho euleriano a partir de s.
* adj[u] = lista de {v, edge_id}: aresta u-v com indice edge_id.
* Para grafo nao-direcionado, adicione cada aresta em ambas as direcoes
* com o mesmo edge_id; o algoritmo marca arestas visitadas pelo id.
*/
vector<int> euler(vector<vector<pair<int,int>>& adj, int s) {
    int E = 0;
    for (auto& v : adj) E += v.size();
    E /= 2; // para nao-direcionado; para direcionado, remova o /2

    vector<bool> used_edge(E, false);
    vector<int> ptr(adj.size(), 0);
    vector<int> path, stk = {s};

    while (!stk.empty()) {
        int v = stk.back();
        bool found = false;
        while (ptr[v] < (int)adj[v].size()) {
            auto [u, eid] = adj[v][ptr[v]++];
            if (!used_edge[eid]) {
                used_edge[eid] = true;
                stk.push_back(u);
                found = true;
                break;
            }
        }
        if (!found) {
            path.push_back(v);
            stk.pop_back();
        }
    }

    reverse(path.begin(), path.end());
    return path;
}

```

Lca

```

11B #include "../data-structures/sparse-table/sparse-table.hpp"

/* LCA (Lowest Common Ancestor) via RMQ
* Encontra o ancestral comum mais proximo de dois vertices em uma arvore.
* Complexidade: O(N log N) no build, O(1) por query.
* Memoria: O(N log N)
* Requisitos:
* - Grafo deve ser uma arvore (conexo e aciclico).
* - Dependencia: sparse-table/sparse-table.hpp
* Metodos:
* - lca(u, v): Retorna o LCA de u e v.
* - dist(u, v): Retorna a distancia entre u e v.
* - is_ancestor(u, v): Verifica se u e ancestral de v.
* - parent(u): Retorna o pai de u.
*/

542 struct LCANode {
430     int dep, id;
89D     LCANode(int d = 1e9, int pos = -1) : dep(d), id(pos) {}

5CF     static inline LCANode merge(const LCANode& l, const LCANode& r) {
DC4         return l.dep < r.dep ? l : r;
505     }
DAD };

33E struct LCA {
060     int N;
41B     vector<int> first, tour, d, p;
73B     SparseTable<LCANode> st;

DCD     LCA(int n, int root, const vector<vector<int>>& adj)
D75         : N(n), first(n), d(n), p(n, -1), st(vector<LCANode>()) {

91C         vector<int> tour_depths;
C8A         tour.reserve(2 * n);
02D         tour_depths.reserve(2 * n);

A12         auto dfs = [&](auto self, int u, int parent_id, int dep) -> void {
73F             p[u] = parent_id;
701             d[u] = dep;
D86             first[u] = tour.size();
FD8             tour.push_back(u);
09A             tour_depths.push_back(dep);
372             for (int v : adj[u]) {
F21                 if (v == parent_id) continue;
207                 self(self, v, u, dep + 1);
FD8                 tour.push_back(u);
09A                 tour_depths.push_back(dep);
FF7             }
0D5         };
7D9         dfs(dfs, root, -1, 0);
9F1         st = SparseTable<LCANode>(tour_depths);
BCA     }

```

```

7B9     int parent(int u) {
F60         return p[u];
47F     }

310     int lca(int u, int v) {
B63         int l = first[u], r = first[v];
A13         if (l > r) swap(l, r);
369         return tour[st.query(l, r).id];
1E6     }

C11     int dist(int u, int v) {
05C         return d[u] + d[v] - 2 * d[lca(u, v)];
465     }

F31     bool is_ancestor(int u, int v) {
645         return lca(u, v) == u;
C22     }
39E };

```

Tarjan

```

/* Tarjan SCC (Strongly Connected Components)
* Complexidade: O(V + E)
*
* Campos publicos apos build():
* comp[v] = id do SCC de v (0-indexado, ordem topologica reversa)
* scc[i] = lista de vertices do i-esimo SCC
* dag = grafo condensado (arestas entre SCCs distintos, sem duplicatas)
* n_scc = numero de SCCs
*
* Nota: comp[] esta em ordem topologica reversa do DAG condensado,
* i.e., se ha aresta SCC a -> SCC b, entao comp[a] > comp[b].
* Para ordem topologica direta, percorra de n_scc-1 a 0.
*/
FA3 struct Tarjan {
058     int n, n_scc = 0;
A41     vector<vector<int>> g, scc, dag;
F7A     vector<int> comp;

770     Tarjan(int n) : n(n), g(n), comp(n, -1) {}

224     Tarjan(const vector<vector<int>>& adj)
35F         : n(adj.size()), g(adj), comp(adj.size(), -1) { build(); }

58B     void add_edge(int u, int v) {
F2B         g[u].push_back(v);
C2B     }

0A8     void build() {
E7C         vector<int> low(n), num(n, -1), stk;
C05         vector<bool> on_stk(n, false);
2DC         int timer = 0;

36D         function<void(int)> dfs = [&](int v) {
006             low[v] = num[v] = timer++;
B00             stk.push_back(v);
48C             on_stk[v] = true;
E32             for (int u : g[v]) {
9CE                 if (num[u] == -1) { dfs(u); low[v] = min(low[v], low[u]); }
52B                 else if (on_stk[u]) low[v] = min(low[v], num[u]);
38B             }
46E             if (low[v] == num[v]) {
861                 scc.push_back({});
667                 while (true) {
D1E                     int u = stk.back(); stk.pop_back();
635                     on_stk[u] = false;
65B                     comp[u] = n_scc;
588                     scc.back().push_back(u);
163                     if (u == v) break;
87F                 }
385                 n_scc++;
6CD             }
5A5         };

830         for (int i = 0; i < n; i++)
917             if (num[i] == -1) dfs(i);

9E4         dag.assign(n_scc, {});
19E         for (int u = 0; u < n; u++)
4FB             for (int v : g[u])
C76                 if (comp[u] != comp[v])
FC7                     dag[comp[u]].push_back(comp[v]);

404         for (auto& adj : dag) {
324             sort(adj.begin(), adj.end());
942             adj.erase(unique(adj.begin(), adj.end()), adj.end());
80E         }
BA2     };
6D7 };

```

Articulations

```

/* Pontos de Articulaçao - Tarjan
* Complexidade: O(V + E)
* Grafo nao-direcionado.
*
* Um vertice e ponto de articulacao se sua remocao desconecta o grafo.

```

```

*
* Campos publicos apos build():
* is_ap[v] = true se v e ponto de articulacao
* get() = lista de vertices que sao pontos de articulacao
*/
E5A struct Articulations {
1A8     int n;
789     vector<vector<int>> g;
BC9     vector<bool> is_ap;

620     Articulations(int n) : n(n), g(n), is_ap(n, false) {}
6F5     Articulations(const vector<vector<int>>& adj)
B9D         : n(adj.size()), g(adj), is_ap(adj.size(), false) { build(); }

58B     void add_edge(int u, int v) {
F2B         g[u].push_back(v);
4D6         g[v].push_back(u);
FFF     }

0A8     void build() {
AC3         vector<int> num(n, -1), low(n);
2DC         int timer = 0;

        function<void(int,int)> dfs = [&](int v, int par) {
006             low[v] = num[v] = timer++;
E07             int children = 0;
E32             for (int u : g[v]) {
403                 if (num[u] == -1) {
C5F                     children++;
412                     dfs(u, v);
C80                     low[v] = min(low[v], low[u]);
369                     if (par == -1 && children > 1) is_ap[v] = true;
1B3                     if (par != -1 && low[u] >= num[v]) is_ap[v] = true;
709                     } else if (u != par) {
2D3                         low[v] = min(low[v], num[u]);
933                     }
9F1                 }
8CA             };

830             for (int i = 0; i < n; i++)
10B                 if (num[i] == -1) dfs(i, -1);
DEE         };

// Retorna lista de vertices que sao pontos de articulacao
D3D     vector<int> get() const {
02F         vector<int> res;
830         for (int i = 0; i < n; i++)
BA3             if (is_ap[i]) res.push_back(i);
B50         return res;
60A     }
A5D };

```

Bridges

```

/* Pontes (Bridges) - Tarjan
* Complexidade: O(V + E)
* Grafo nao-direcionado.
*
* Uma aresta e ponte se sua remocao desconecta o grafo.
*
* Campos publicos apos build():
* bridges = lista de arestas {u, v} que sao pontes
*/
D44 struct Bridges {
1A8     int n;
789     vector<vector<int>> g;
0CC     vector<pair<int,int>> bridges;

B0A     Bridges(int n) : n(n), g(n) {}
C65     Bridges(const vector<vector<int>>& adj)
5F7         : n(adj.size()), g(adj) { build(); }

58B     void add_edge(int u, int v) {
F2B         g[u].push_back(v);
4D6         g[v].push_back(u);
FFF     }

0A8     void build() {
AC3         vector<int> num(n, -1), low(n);
2DC         int timer = 0;

CF3         function<void(int,int)> dfs = [&](int v, int par) {
006             low[v] = num[v] = timer++;
E32             for (int u : g[v]) {
403                 if (num[u] == -1) {
412                     dfs(u, v);
C80                     low[v] = min(low[v], low[u]);
2FE                     if (low[u] > num[v]) bridges.push_back({v, u});
FB4                 } else if (u != par) {
2D3                     low[v] = min(low[v], num[u]);
E66                 }
7A3             }
938         };

830         for (int i = 0; i < n; i++)
10B             if (num[i] == -1) dfs(i, -1);

```

```
A6F    };
000 };
```

2-SAT

```
C31 #include "tarjan.hpp"
```

```
/* 2-SAT - Complexidade: O(V + E)
 *
 * Literal x: use i para x_i=true, ~i para x_i=false.
 *
 * add_impl(x, y)  -> x -> y (contrapositiva ~y -> ~x automatica)
 * add_or(x, y)    -> x v y
 * add_xor(x, y)   -> exatamente um de x, y e true
 * add_iff(x, y)   -> x = y
 * force(x)        -> x = true
 * solve()         -> {satisfativo, ans[]}; ans[i] = 0 ou 1
 */

D9D struct TwoSat {
1A8     int n;
0D7     Tarjan t;
DAB     vector<int> ans;

46B     TwoSat(int n) : n(n), t(2 * n), ans(n, -1) {}

    // converte literal para no interno: i >= 0 + 2*i, i < 0 + -2*i-1
5D7     static int node(int x) { return x >= 0 ? 2 * x : -2 * x - 1; }

    // implicacao x -> y (e contrapositiva ~y -> ~x)
74A     void add_impl(int x, int y) {
D64         int nx = node(x), ny = node(y);
2B2         t.add_edge(nx, ny);
378         t.add_edge(ny ^ 1, nx ^ 1);
255     }

    // x v y
F10     void add_or(int x, int y) { add_impl(~x, y); }

    // x XOR y (exatamente um verdadeiro)
C02     void add_xor(int x, int y) { add_or(x, y); add_or(~x, ~y); }

    // x = y
721     void add_iff(int x, int y) { add_impl(x, y); add_impl(~x, ~y); }

    // forza x = true
5F8     void force(int x) { add_impl(~x, x); }

A8E     pair<bool, vector<int>> solve() {
2BD         t.build();
E1C         ans.assign(n, -1);
603         for (int i = 0; i < n; i++) {
EBA             if (t.comp[2 * i] == t.comp[2 * i + 1]) return {false, {}};
7B4             ans[i] = t.comp[2 * i] > t.comp[2 * i + 1] ? 1 : 0;
291         }
997         return {true, ans};
BB1     }
F3D };
```

Hld Commutative

```
6A0 #include "../data-structures/segment-tree/segment-tree.hpp"
```

```
/* HLD Comutativo (Update Pontual)
 * Para operacoes onde A + B = B + A.
 * Requisito: data-structures/segment-tree/segment-tree.hpp
 */

8A0 template<typename NODE>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
115     SegTree<NODE> st;

7EC     HLD(const vector<vector<int>>& adj, const vector<NODE>& vals, int root = 0)
833         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n), st(n) {}

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
ABE             d[v] = d[u] + 1; p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
}

330     auto dfs_hld = [&](auto self, int u) -> void {
408         pos[u] = t++;
6EA         tour[pos[u]] = u;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
head[v] = (i == 0 ? head[u] : v);
self(self, v);
}

dfs_sz(dfs_sz, root);
head[root] = root;
dfs_hld(dfs_hld, root);
```

```
07D     }
A11 };

89B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
113     };

A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);

759     vector<NODE> base(n);
830     for(int i = 0; i < n; i++)
101         base[pos[i]] = vals[i];
E4E     st = SegTree<NODE>(base);
DE8 }

E1B     void update(int u, const NODE& val) {
C7A         st.update(pos[u], val);
080     }

0F8     NODE query_path(int u, int v) {
ADA         NODE res;
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
A0D             res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025             v = p[head[v]];
41F         }
0E0         if (d[u] > d[v]) swap(u, v);
DC7         return NODE::merge(res, st.query(pos[u], pos[v]));
81B     }

51A     NODE query_subtree(int u) {
0CF         return st.query(pos[u], pos[u] + sz[u] - 1);
747     }
058 };
```

Hld Commutative Lazy

```
2F0 #include "../data-structures/segment-tree/lazy-segment-tree.hpp"
```

```
/* HLD Comutativo com Lazy Propagation
 * Para operacoes onde A + B = B + A.
 * Permite updates e queries em caminhos e subarvores.
 * Requisito: data-structures/segment-tree/lazy-segment-tree.hpp
 */

708 template<typename NODE, typename TAG>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
FFF     LazySegmentTree<NODE, TAG> st;

7EC     HLD(const vector<vector<int>>& adj, const vector<NODE>& vals, int root = 0)
833         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n), st(n) {}

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
ABE             d[v] = d[u] + 1; p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
D9D             if (sz[v] > max_sz) { max_sz = sz[v]; best_v = v; }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
}

330     auto dfs_hld = [&](auto self, int u) -> void {
408         pos[u] = t++;
6EA         tour[pos[u]] = u;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
head[v] = (i == 0 ? head[u] : v);
self(self, v);
}

dfs_sz(dfs_sz, root);
head[root] = root;
dfs_hld(dfs_hld, root);
```

```
759     vector<NODE> base(n);
830     for(int i = 0; i < n; i++)
101         base[pos[i]] = vals[i];
9F8     st = LazySegmentTree<NODE, TAG>(base);
188 }

D7C     void update_path(int u, int v, const TAG& tag) {
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
EC8             st.update(pos[head[v]], pos[v], tag);
025             v = p[head[v]];
DD0         }
0E0         if (d[u] > d[v]) swap(u, v);
07A         st.update(pos[u], pos[v], tag);
656     }

0F8     NODE query_path(int u, int v) {
ADA         NODE res;
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
A0D             res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025             v = p[head[v]];
41F         }
0E0         if (d[u] > d[v]) swap(u, v);
DC7         return NODE::merge(res, st.query(pos[u], pos[v]));
81B     }

F81     void update_subtree(int u, const TAG& tag) {
675         st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D     }

51A     NODE query_subtree(int u) {
0CF         return st.query(pos[u], pos[u] + sz[u] - 1);
747     }
AA5 };
```

Hld Non Commutative

```
6A0 #include "../data-structures/segment-tree/segment-tree.hpp"
```

```
/* HLD Nao-Comutativo (Update Pontual)
 * Para operacoes onde A * B != B * A.
 * Requisito: DoubleNode para gerenciar merges em ambas as direcoes.
 */

8A0 template<typename NODE>
0FF struct DoubleNode {
A30     NODE down, up;
F17     DoubleNode() : down(NODE()), up(NODE()) {}
77F     DoubleNode(const NODE& n) : down(n), up(n) {}
31A     DoubleNode(const NODE& d, const NODE& u) : down(d), up(u) {}

CFE     static inline DoubleNode merge(const DoubleNode& l, const DoubleNode& r) {
2D1         return {
15C             NODE::merge(l.down, r.down),
F17             NODE::merge(r.up, l.up)
5D3         };
809     };
3DC };

8A0 template<typename NODE>
403 struct HLD {
577     int n, t;
256     vector<int> p, sz, d, head, pos, tour;
58B     SegTree<DoubleNode<NODE>> st;

7EC     HLD(const vector<vector<int>>& adj, const vector<NODE>& vals, int root = 0)
7F8         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n), tour(n), st(n) {}

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
ABE             d[v] = d[u] + 1; p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
D9D             if (sz[v] > max_sz) { max_sz = sz[v]; best_v = v; }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
}

330     auto dfs_hld = [&](auto self, int u) -> void {
408         pos[u] = t++;
6EA         tour[pos[u]] = u;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
```

```

BD7         head[v] = (i == 0 ? head[u] : v);
033         self(self, v);
D08     }
54A     };
A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);
9FC     vector<DoubleNode<NODE>> base(n);
830     for(int i = 0; i < n; i++)
9A8         base[pos[i]] = DoubleNode<NODE>(vals[i]);
19C     st = SegTree<DoubleNode<NODE>>(base);
997 }

E1B void update(int u, const NODE& val) {
353     st.update(pos[u], DoubleNode<NODE>(val));
443 }

0F8 NODE query_path(int u, int v) {
34B     NODE L, R;
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
0F3             L = NODE::merge(L, st.query(pos[head[u]], pos[u]).up);
139             u = p[head[u]];
3B2         } else {
B5C             R = NODE::merge(st.query(pos[head[v]], pos[v]).down, R);
025             v = p[head[v]];
7D9         }
DBB     }
6CE     if (d[u] > d[v]) L = NODE::merge(L, st.query(pos[v], pos[u]).up);
B9A     else R = NODE::merge(st.query(pos[u], pos[v]).down, R);
85E     return NODE::merge(L, R);
19F }

51A NODE query_subtree(int u) {
DE6     return st.query(pos[u], pos[u] + sz[u] - 1).down;
588 }
98B };

```

HLD Non Commutative Lazy

```

2F0 #include "../data-structures/segment-tree/lazy-segment-tree.hpp"
/* HLD Nao-Comutativo com Lazy Propagation
 * Para operacoes onde A + B != B + A (ex: Matrices, Funcoes Lineares).
 * Requisito: DoubleNode que guarda versoes 'normal' e 'invertida' do merge.
 */

8A0 template<typename NODE>
0FF struct DoubleNode {
A30     NODE down, up;
F17     DoubleNode() : down(NODE()), up(NODE()) {}
77F     DoubleNode(const NODE& n) : down(n), up(n) {}
31A     DoubleNode(const NODE& d, const NODE& u) : down(d), up(u) {}
CFE     static inline DoubleNode merge(const DoubleNode& l, const DoubleNode& r) {
2D1         return {
15C             NODE::merge(l.down, r.down),
F17             NODE::merge(r.up, l.up)
5D3         };
809     };
BBF     template<typename TAG>
3CC     void apply(const TAG& t, int l, int r) {
B08         down.apply(t, l, r);
2BF         up.apply(t, l, r);
8FB     }
B7B };

708 template<typename NODE, typename TAG>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
C94     LazySegmentTree<DoubleNode<NODE>, TAG> st;
7EC     HLD(const vector<vector<int>>& adj, const vector<NODE>& vals, int root = 0)
833         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n), st(n) {
898         vector<vector<int>> g = adj;
227         d[root] = 0, p[root] = root;
94B         auto dfs_sz = [&](auto self, int u) -> void {
267             sz[u] = 1;
839             int best_v = -1, max_sz = -1;
A8C             for (auto& v : g[u]) {
567                 if (v == p[u]) continue;
ABE                 d[v] = d[u] + 1; p[v] = u;
033                 self(self, v);
CC3                 sz[u] += sz[v];
D9D                 if (sz[v] > max_sz) { max_sz = sz[v]; best_v = v; }
A7E             }
2D5             if (best_v != -1) {
BB6                 for (int i = 0; i < (int)g[u].size(); i++) {
F10                     if (g[u][i] == best_v && i != 0) {
D76                         swap(g[u][0], g[u][i]);
C2B                         break;
27A                     }
}

```

```

CAE     }
07D     }
A11     };
B9B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
113     };
A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);
9FC     vector<DoubleNode<NODE>> base(n);
B89     for(int i = 0; i < n; i++) base[pos[i]] = DoubleNode<NODE>(vals[i]);
54E     st = LazySegmentTree<DoubleNode<NODE>, TAG>(base);
B60 }

D7C void update_path(int u, int v, const TAG& tag) {
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
D76             st.update(pos[head[u]], pos[u], tag);
139             u = p[head[u]];
D79         } else {
EC8             st.update(pos[head[v]], pos[v], tag);
025             v = p[head[v]];
E8B         }
15B     }
B83     if (d[u] > d[v]) st.update(pos[v], pos[u], tag);
052     else st.update(pos[u], pos[v], tag);
855 }

0F8 NODE query_path(int u, int v) {
34B     NODE L, R;
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
0F3             L = NODE::merge(L, st.query(pos[head[u]], pos[u]).up);
139             u = p[head[u]];
3B2         } else {
B5C             R = NODE::merge(st.query(pos[head[v]], pos[v]).down, R);
025             v = p[head[v]];
7D9         }
DBB     }
152     if (d[u] > d[v]) {
7E5         L = NODE::merge(L, st.query(pos[v], pos[u]).up);
D6C     } else {
806         R = NODE::merge(st.query(pos[u], pos[v]).down, R);
8A1     }
85E     return NODE::merge(L, R);
EB1 }

F81 void update_subtree(int u, const TAG& tag) {
675     st.update(pos[u], pos[u] + sz[u] - 1, tag);
2D0 }

51A NODE query_subtree(int u) {
DE6     return st.query(pos[u], pos[u] + sz[u] - 1).down;
588 }
068 };

```

Dinic

```

/* Dinic (Max Flow)
 * Complexidade: O(V^2 * E) geral, O(E * sqrt(V)) em grafos unitarios.
 * Memoria: O(V + E)
 *
 * Metodos:
 *   add_edge(u, v, cap)      → aresta u-v com capacidade cap
 *   add_edge(u, v, cap, rcap) → idem com aresta reversa rcap (nao-direcionado)
 *   flow(s, t)              → fluxo maximo de s a t
 *   min_cut(s)              → apos flow(), vetor onde [i]=true se i esta no lado da fonte
 */

67A template<typename T>
14D struct Dinic {
E9B     struct Edge {
C63         int to, rev;
4AE         T cap;
001     };
060     int N;
ED3     vector<vector<Edge>> g;
538     vector<int> level, iter;
241     Dinic(int n) : N(n), g(n), level(n), iter(n) {}
D8F     void add_edge(int u, int v, T cap, T rcap = 0) {
25C         g[u].push_back({v, (int)g[v].size(), cap});
ACA         g[v].push_back({u, (int)g[u].size() - 1, rcap});
05B     }
123     bool bfs(int s, int t) {
224         fill(level.begin(), level.end(), -1);

```

```

26A     queue<int> q;
EC8     level[s] = 0;
08B     q.push(s);
14D     while (!q.empty()) {
0E6         int v = q.front(); q.pop();
24E         for (auto& e : g[v])
466             if (e.cap > 0 && level[e.to] < 0) {
BD7                 level[e.to] = level[v] + 1;
6F4                 q.push(e.to);
A33             }
FF9         }
FCB     return level[t] >= 0;
56A }

D20 T dfs(int v, int t, T f) {
704     if (v == t) return f;
6E7     for (int& i = iter[v]; i < (int)g[v].size(); i++) {
63C         Edge& e = g[v][i];
2C2         if (e.cap > 0 && level[v] < level[e.to]) {
054             T d = dfs(e.to, t, min(f, e.cap));
1A3             if (d > 0) {
8FC                 e.cap -= d;
772                 g[e.to][e.rev].cap += d;
BE2                 return d;
18D             }
57D         }
E13     }
BB3     return 0;
182 }

903 T flow(int s, int t) {
19A     T res = 0;
8CE     while (bfs(s, t)) {
736         fill(iter.begin(), iter.end(), 0);
A96         T d;
F4F         while ((d = dfs(s, t, numeric_limits<T>::max())) > 0) 12345668900-123
337             res += d;
697     }
B50     return res;
1D8 }

8B0 vector<bool> min_cut(int s) {
F71     vector<bool> vis(N, false);
26A     queue<int> q;
451     vis[s] = true;
08B     q.push(s);
14D     while (!q.empty()) {
0E6         int v = q.front(); q.pop();
24E         for (auto& e : g[v])
F04             if (e.cap > 0 && !vis[e.to]) {
1E8                 vis[e.to] = true;
A56                 q.push(e.to);
607             }
C81     return vis;
71F }
CB5 };

```

MCMF

```

/* MCMF – Min-Cost Max-Flow (SPFA/Bellman-Ford based)
 * Complexidade: O(V * E * flow) no pior caso; na pratica muito mais rapido.
 * Memoria: O(V + E)
 *
 * Metodos:
 *   add_edge(u, v, cap, cost)      → aresta u-v com cap e custo
 *   add_edge(u, v, cap, rcap, cost) → idem com aresta reversa rcap
 *   flow(s, t)                    → {fluxo maximo, custo minimo} de s a t
 *   flow(s, t, max_flow)          → idem limitando o fluxo total a max_flow
 *
 * Observacoes:
 *   - Suporta custos negativos (usa SPFA em vez de Dijkstra).
 *   - Sem custos negativos, prefira Dijkstra+potenciais: O(E log V)/aug.
 *   - 0 custo retornado e o custo minimo para enviar o fluxo maximo.
 */

39C template<typename Cap, typename Cost>
6F3 struct MCMF {
E9B     struct Edge {
C63         int to, rev;
2E2         Cap cap;
CB9         Cost cost;
BFF     };
060     int N;
ED3     vector<vector<Edge>> g;
7E6     MCMF(int n) : N(n), g(n) {}
251     void add_edge(int u, int v, Cap cap, Cost cost, Cap rcap = 0) {
C5D         g[u].push_back({v, (int)g[v].size(), cap, cost});
D90         g[v].push_back({u, (int)g[u].size() - 1, rcap, -cost});
200     }
C97     pair<Cap, Cost> flow(int s, int t,
25F         Cap max_flow = numeric_limits<Cap>::max()) {

```

```

8AC      Cap total_flow = 0;
8F8      Cost total_cost = 0;

BF6      while (max_flow > 0) {
715          vector<Cost> dist(N, numeric_limits<Cost>::max());
7A2          vector<bool> in_queue(N, false);
FF9          vector<int> prev_node(N, -1), prev_edge(N, -1);

A93          dist[s] = 0;
26A          queue<int> q;
08B          q.push(s);
6FD          in_queue[s] = true;

14D          while (!q.empty()) {
0E6              int v = q.front(); q.pop();
F19              in_queue[v] = false;
775              for (int i = 0; i < (int)g[v].size(); i++) {
027                  auto& e = g[v][i];
AB4                  if (e.cap > 0 && dist[v] + e.cost < dist[e.to]) {
9B1                      dist[e.to] = dist[v] + e.cost;
3C0                      prev_node[e.to] = v;
2E2                      prev_edge[e.to] = i;
5CA                      if (!in_queue[e.to]) {
304                          in_queue[e.to] = true;
6F4                          q.push(e.to);
8A6                      }
5E7                  }
168              }
A8B          }

472          if (dist[t] == numeric_limits<Cost>::max()) break;
1E4          Cap pushed = max_flow;
8CE          for (int v = t; v != s; v = prev_node[v])
31C              pushed = min(pushed, g[prev_node[v]][prev_edge[v]].cap);

879          for (int v = t; v != s; v = prev_node[v]) {
49B              auto& e = g[prev_node[v]][prev_edge[v]];
A04              e.cap -= pushed;
B9A              g[v][e.rev].cap += pushed;
F38          }

3BD          total_flow += pushed;
6CE          total_cost += pushed * dist[t];
12C          max_flow -= pushed;
CC5      }

570      return {total_flow, total_cost};
D1A  }
0B4  };

```

Bipartite Matching

```

0AA  #include "dinic.hpp"

/* Bipartite Matching via Dinic
 * Matching maximo em grafo bipartido.
 * Complexidade: O(E * sqrt(V))
 *
 * Vertices esquerda: [0, n), direita: [n, n+m).
 *
 * Campos publicos apos match():
 * size = tamanho do matching maximo
 * match_l[i] = vertice da direita matched com i (-1 se nao matched)
 * match_r[j] = vertice da esquerda matched com j (-1 se nao matched)
 *
 * Metodos:
 * add_edge(i, j) -> aresta entre esquerda i e direita j
 * match() -> executa o matching, retorna tamanho
 */

878 struct BipartiteMatching {
14E     int n, m;
690     Dinic<int> g;
3AE     int S, T;
2AF     vector<int> match_l, match_r;
689     int size = 0;

1FD     BipartiteMatching(int n, int m)
104         : n(n), m(m), g(n + m + 2),
75F         S(n + m), T(n + m + 1),
779         match_l(n, -1), match_r(m, -1) {
1D3         for (int i = 0; i < n; i++) g.add_edge(S, i, 1);
CC1         for (int j = 0; j < m; j++) g.add_edge(n + j, T, 1);
B60     }

C5E     void add_edge(int i, int j) {
3EC         g.add_edge(i, n + j, 1);
EE4     }

274     int match() {
5D7         size = g.flow(S, T);
830         for (int i = 0; i < n; i++)
B83             for (auto& e : g.g[i])
772                 if (e.to != S && e.cap == 0) {
846                     match_l[i] = e.to - n;
78E                     match_r[e.to - n] = i;
FE7                 }

```

```

1C6         return size;
B47     }
CFA };

```

Math

Mint

```

/* mint - Inteiro modular
 * Complexidade: O(1) por operacao, O(log mod) para inv()
 *
 * template<typename T, T mod>
 *
 * Uso comum:
 * using mint = Mint<int, 998244353>;
 * using mll = Mint<ll, 998244353LL>;
 *
 * Operacoes: +, -, *, /, ==, !=, <, <=
 * inv() - inverso modular (mod deve ser primo)
 */

2DB template<typename T, T mod>
4D0 struct Mint {
CA0     T v;
926     Mint(T v = 0) : v(((v % mod) + mod) % mod) {}

E93     Mint operator+(Mint o) const { return v+o.v >= mod ? v+o.v-mod : v+o.v; }
66C     Mint operator-(Mint o) const { return v-o.v < 0 ? v-o.v+mod : v-o.v; }
3B1     Mint operator*(Mint o) const { return (ll)v * o.v % mod; }
5B6     Mint operator/(Mint o) const { return *this * o.inv(); }

96C     Mint& operator+=(Mint o) { return *this = *this + o; }
BB0     Mint& operator-=(Mint o) { return *this = *this - o; }
926     Mint& operator*=(Mint o) { return *this = *this * o; }
E1C     Mint& operator/=(Mint o) { return *this = *this / o; }

8D1     bool operator==(Mint o) const { return v == o.v; }
5B7     bool operator!=(Mint o) const { return v != o.v; }

A45     explicit operator T() const { return v; }

EC5     Mint inv() const {
FD5         T a = v, b = mod, x = 1, y = 0;
367         while (b) { T q = a/b; a -= q*b; swap(a,b); x -= q*y; swap(x,y); }
EA5         return x;
0CF     }

09A     friend ostream& operator<<(ostream& os, Mint m) { return os << m.v; }
AAA };

```

Fast Exponentiation

```

/* fexp - Exponenciacao rapida
 * Complexidade: O(log b)
 *
 * fexp(a, b) -> a^b usando operator* do tipo T
 *
 * Use com mint/mll para exponenciacao modular automatica.
 *
 * modinv<mod>(a) -> a^-1 % mod (mod deve ser primo)
 */

67A template<typename T>
83D T fexp(T a, ll b) {
EC7     T res = T(1);
0EB     for (; b > 0; b >= 1, a = a*a)
FC2         if (b & 1) res = res * a;
B50     return res;
0CE }

31E template<ll mod>
611 ll modinv(ll a) { return fexp<ll>{a % mod, mod - 2}; }

```

Prime Sieve

```

/* Crivo de Eratostenes
 * Complexidade: O(N log log N)
 *
 * sieve(n) - vetor com todos os primos ate n (inclusive).
 */

705 vector<int> sieve(int n) {
EFA     vector<bool> is_prime(n + 1, true);
FD3     vector<int> primes;
19E     is_prime[0] = is_prime[1] = false;
8A9     for (int p = 2; p <= n; p++) {
29C         if (is_prime[p]) {
BCE             primes.push_back(p);
908             if ((llong long)p * p <= n)
4C7                 for (int i = p * p; i <= n; i += p)
F10                 is_prime[i] = false;
FFE         }
5D1     }
A20     return primes;
81C }

```

Euler Phi

```

/* Crivo Linear - O(N)
 *
 * linear_sieve(n) -> {primes, phi}
 * primes : todos os primos ate n (inclusive)
 * phi : phi[i] = φ(i) para i em [0, n]
 */

9A7 pair<vector<int>, vector<int>> linear_sieve(int n) {
303     vector<int> phi(n + 1, 0), primes;
614     vector<bool> composite(n + 1, false);
678     phi[1] = 1;
508     for (int i = 2; i <= n; i++) {
C54         if (!composite[i]) {
E74             primes.push_back(i);
ABD             phi[i] = i - 1;
6DB         }
3B9         for (int p : primes) {
0E8             if ((llong long)i * p > n) break;
847             composite[i * p] = true;
569             if (i % p == 0) {
8EC                 phi[i * p] = phi[i] * p;
C2B                 break;
792             } else {
812                 phi[i * p] = phi[i] * (p - 1);
988             }
A0F         }
8CB     }
6EF     return {primes, phi};
158 }

```

Number Theory

```

/* GCD / LCM - O(log min(a, b)) */

225 ll gcd(ll a, ll b) { return b ? gcd(b, a % b) : a; }
15C ll lcm(ll a, ll b) { return a / gcd(a, b) * b; }

/* Divisores de n - O(sqrt(N))
 * Retorna lista de divisores em ordem crescente.
 */

5FA vector<ll> divisors(ll n) {
17E     vector<ll> lo, hi;
148     for (ll i = 1; i * i <= n; i++) {
C06         if (n % i == 0) {
1EB             lo.push_back(i);
9C3             if (i != n / i) hi.push_back(n / i);
B5B         }
2DE     }
FC1     reverse(hi.begin(), hi.end());
3AF     lo.insert(lo.end(), hi.begin(), hi.end());
253     return lo;
B9C }

/* Fatoracao em primos - O(sqrt(N))
 * Retorna lista de {primo, expoente} em ordem crescente.
 */

7E3 vector<pair<ll,int>> factorize(ll n) {
618     vector<pair<ll,int>> factors;
D2E     for (ll p = 2; p * p <= n; p++) {
80A         if (n % p == 0) {
A01             int exp = 0;
6C9             while (n % p == 0) { n /= p; exp++; }
13A             factors.push_back({p, exp});
2E9         }
47D     }
992     if (n > 1) factors.push_back({n, 1});
FA9     return factors;
92A }

```

```

/* CRT - Teorema Chines dos Restos - O(log min(m1, m2))
 * Retorna {x, lcm(m1,m2)} tal que x = a1 (mod m1) e x = a2 (mod m2).
 * Retorna {0, 0} se nao ha solucao (m1, m2 nao coprimos e incompativeis).
 */

8A9 pair<ll,ll> crt(ll a1, ll m1, ll a2, ll m2) {
8F7     ll g = gcd(m1, m2);
916     if ((a2 - a1) % g != 0) return {0, 0};
9C3     ll l = m1 / g * m2;
D0B     ll a = m1/g, b = m2/g, x = 1, y = 0;
E58     for (ll ta = a, tb = b; tb; ) {
146         ll q = ta/tb; ta -= q*tb; swap(ta,tb); x -= q*y; swap(x,y);
54E     }
581     x = (x % (m2/g) + m2/g) % (m2/g);
EE5     ll r = (a1 + m1 * ((a2 - a1) / g % (m2/g) * x % (m2/g))) % l;
D27     return {(r + l) % l, l};
F7F }

/* Fatoracao usando lista de primos pre-computada - O(n(sqrt(N)))
 * Use quando ja tiver rodado sieve(sqrt(n)) antes de chamar para varios n.
 * primos deve conter todos os primos ate sqrt(n).
 */

```



```
E6D vector<pair<ll,int>> factorize(ll n, const vector<int>& primes) {
618 vector<pair<ll,int>> factors;
3B9 for (int p : primes) {
851 if ((ll)p * p > n) break;
80A if (n % p == 0) {
A01 int exp = 0;
6C9 while (n % p == 0) { n /= p; exp++; }
13A factors.push_back({p, exp});
2E9 }
17C }
992 if (n > 1) factors.push_back({n, 1});
FA9 return factors;
490 }
```

Miller Rabin

```
/* Miller-Rabin + Pollard Rho
*
* is_prime(n) → 0(log² n), determinístico para n < 3.3·10¹⁸
* factorize(n) → 0(n^{1/4} log n) esperado; fatores com repeticao
*
* Use sort + unique para fatores distintos, ou agrupe por expoente.
*/
7D4 ll mulmod(ll a, ll b, ll m) { return (__int128)a * b % m; }
0A0 bool is_prime(ll n) {
5A7 if (n < 2) return false;
0A8 if (n == 2 || n == 3 || n == 5 || n == 7) return true;
3CE if (n % 2 == 0 || n % 3 == 0) return false;
B87 ll d = n - 1; int r = 0;
5B7 while (d % 2 == 0) d /= 2, r++;
C12 for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
D22 if (a >= n) continue
8C8 ll x = 1, base = a % n, exp = d;
006 for (; exp > 0; exp >>= 1, base = mulmod(base, base, n))
5E6 if (exp & 1) x = mulmod(x, base, n);
8A3 if (x == 1 || x == n - 1) continue;
30D bool composite = true;
21B for (int i = 0; i < r - 1; i++) {
F38 x = mulmod(x, x, n);
789 if (x == n - 1) { composite = false; break; }
4C7 }
4F2 if (composite) return false;
D5C }
8A6 return true;
E78 }
711 ll pollard_rho(ll n) {
55A if (n % 2 == 0) return 2;
2BC ll x = rand() % (n - 2) + 2, y = x, c = rand() % (n - 1) + 1, d = 1;
4EB while (d == 1) {
C17 x = (mulmod(x, x, n) + c) % n;
4EE y = (mulmod(y, y, n) + c) % n;
4EE y = (mulmod(y, y, n) + c) % n;
DD8 d = __gcd(abs(x - y), n);
565 }
9EC return d == n ? pollard_rho(n) : d;
5D7 }
6C2 vector<ll> factorize(ll n) {
1B9 if (n == 1) return {};
970 if (is_prime(n)) return {n};
B32 ll d = pollard_rho(n);
F9F auto a = factorize(d), b = factorize(n / d);
19E a.insert(a.end(), b.begin(), b.end());
3F5 return a;
E13 }
```

Matrix

```
AD1 #include "fexp.hpp"
```

```
/* Matrix – Matriz quadrada com multiplicacao e exponenciacao rapida
* Complexidade: 0(n³ log k) para M^k
*
* Use com mint/mll para operacoes modulares automaticas.
*
* Matrix<T> M(n) → identidade nxn
* Matrix<T> M(n,0) → matriz nxn zerada
* M[i][j] → acesso direto
* A * B → multiplicacao 0(n³)
* M ^ k → M^k usando fexp
*/
```

```
67A template<typename T>
BF3 struct Matrix {
1A8 int n;
C24 vector<vector<T>> mat;
C62 Matrix(int n, int identity = 1) : n(n), mat(n, vector<T>(n, T(0))) {
F03 if (identity) for (int i = 0; i < n; i++) mat[i][i] = T(1);
68A }
90D vector<T>& operator[](int i) { return mat[i]; }
699 const vector<T>& operator[](int i) const { return mat[i]; }
```

```
AFB Matrix operator*(const Matrix& rhs) const {
83B Matrix res(n, 0);
830 for (int i = 0; i < n; i++)
23D for (int k = 0; k < n; k++) if (mat[i][k] != T(0))
F90 for (int j = 0; j < n; j++)
C55 res[i][j] = res[i][j] + mat[i][k] * rhs.mat[k][j];
B50 return res;
893 }
6FC Matrix operator^(long long b) const { return fexp(*this, b); }
732 };
```

Gaussian Elimination

```
/* Eliminacao Gaussiana (RREF) – 0(N² * M)
* Funciona com double ou Mint<T, mod>.
*
* gauss(A, b) → resolve Ax = b.
* Retorna {x, true} se unica, {x, false} se inconsistente/sub-determinado.
*/
67A template<typename T>
5D7 pair<vector<T>, bool> gauss(vector<vector<T>> A, vector<T> b) {
FE3 int n = A.size();
3E0 for (int i = 0; i < n; i++) A[i].push_back(b[i]);
3D7 int m = n + 1, rank = 0;
079 for (int col = 0; col < n && rank < n; col++) {
9D3 int sel = rank;
F97 for (int i = rank + 1; i < n; i++)
E75 if (abs(A[i][col]) > abs(A[sel][col])) sel = i;
9E6 if (!(bool)A[sel][col]) continue;
D8D swap(A[sel], A[rank]);
B46 T iv = T(1) / A[rank][col];
C24 for (int j = col; j < m; j++) A[rank][j] = A[rank][j] * iv;
603 for (int i = 0; i < n; i++) {
BC7 if (i == rank || !(bool)A[i][col]) continue;
BEF T f = A[i][col];
BAC for (int j = col; j < m; j++) A[i][j] = A[i][j] - f * A[rank][j];
371 }
56E rank++;
3F2 }
175 if (rank < n) return {{}, false};
A1B vector<T> x(n);
B95 for (int i = 0; i < n; i++) x[i] = A[i].back();
079 return {x, true};
724 }
```

Linear Basis

```
/* Linear Basis (Base Linear XOR)
* Complexidade: 0(B) por operacao, onde B = numero de bits (64 para ll)
* Memoria: 0(B)
*
* Metodos:
* insert(x) → insere x; retorna true se x e linearmente independente
* contains(x) → true se x e XOR de algum subconjunto da base
* max_xor(x=0) → maior XOR de x com qualquer subconjunto da base
* min_xor() → menor XOR nao-nulo possivel (0 se base incompleta)
* size() → numero de elementos linearmente independentes
* merge(other) → une duas bases (insere todos os elementos de other)
*
* Observacoes:
* - Base em forma reduzida (cada bit pivot aparece em apenas uma linha).
* - Para k-esimo XOR crescente, use reduce() e trate como numero binario.
*/
```

```
538 template<typename T = ll, int B = 63>
F45 struct LinearBasis {
E9C array<T, B + 1> basis{};
1FC int sz = 0;
220 bool insert(T x) {
C8C for (int i = B; i >= 0; i--) {
760 if (!(x >> i) & 1) continue;
617 if (!basis[i]) {
C6C basis[i] = x;
EB9 sz++;
8A6 return true;
50C }
6B0 x ^= basis[i];
8A2 }
D1F return false;
CD7 }
394 bool contains(T x) const {
C8C for (int i = B; i >= 0; i--) {
760 if (!(x >> i) & 1) continue;
260 if (basis[i]) return false;
6B0 x ^= basis[i];
1A9 }
8A6 return true;
189 }
C3B T max_xor(T x = 0) const {
```

```
086 for (int i = B; i >= 0; i--)
EC9 x = max(x, x ^ basis[i]);
EA5 return x;
74B }
BBB T min_xor() const {
F56 for (int i = 0; i <= B; i++)
15D if (basis[i]) return basis[i];
BB3 return 0;
5FC }
21A int size() const { return sz; }
AC0 void merge(const LinearBasis& other) {
086 for (int i = B; i >= 0; i--)
D67 if (other.basis[i]) insert(other.basis[i]);
1CA }
```

```
/* Reduz a base para forma escalonada reduzida (cada pivot tem 1 bit).
* Necessario para consultas de k-esimo XOR.
* Apos reduce(), qualquer subconjunto gera um XOR distinto:
* e possivel enumerar todos os 2^sz XORs possiveis.
*/
```

```
void reduce() {
93A for (int i = B; i >= 0; i--) {
C8C if (!basis[i]) continue;
F14 for (int j = i + 1; j <= B; j++)
E36 if ((basis[j] >> i) & 1) basis[j] ^= basis[i];
7E7 }
CA5 }
563 }
/* k-esimo menor XOR (0-indexado) entre todos os 2^sz subconjuntos.
* Requer reduce() antes. Se sz < B+1, XOR=0 = subconjunto vazio.
* k=0 → 0; k=1 → menor XOR nao-nulo, etc.
*/
0B4 T kth(ll k) const {
E55 vector<T> elems;
F56 for (int i = 0; i <= B; i++)
210 if (basis[i]) elems.push_back(basis[i]);
// elems esta em ordem crescente de bit mais significativo
if (k >= (1LL << (int)elems.size())) return -1; // k fora do range
T res = 0;
687 for (int i = 0; i < (int)elems.size(); i++)
A7C if ((k >> i) & 1) res ^= elems[i];
B50 return res;
B1A }
DE8 };
```

Unimodal Search

```
/* Busca unimodal – f deve ser unimodal no intervalo dado.
*
* ternary_min/max(lo, hi, f) → inteiros, 0(log N)
* golden_min/max(lo, hi, f, iters=200) → continuo, 0(iters); 200 iters = 1e-30
*/
```

```
// Ternary search para minimo em inteiros [lo, hi]
398 template<typename F>
209 long long ternary_min(long long lo, long long hi, F f) {
960 while (hi - lo > 2) {
C9D long long m1 = lo + (hi - lo) / 3;
A33 long long m2 = hi - (hi - lo) / 3;
2EE if (f(m1) <= f(m2)) hi = m2;
F67 else lo = m1;
885 }
52F long long best = lo;
35B for (long long x = lo + 1; x <= hi; x++)
EB8 if (f(x) < f(best)) best = x;
F26 return best;
DB5 }
```

```
// Ternary search para maximo em inteiros [lo, hi]
398 template<typename F>
E4E long long ternary_max(long long lo, long long hi, F f) {
960 while (hi - lo > 2) {
C9D long long m1 = lo + (hi - lo) / 3;
A33 long long m2 = hi - (hi - lo) / 3;
750 if (f(m1) >= f(m2)) hi = m2;
F67 else lo = m1;
344 }
52F long long best = lo;
35B for (long long x = lo + 1; x <= hi; x++)
592 if (f(x) > f(best)) best = x;
F26 return best;
2AD }
```

```
// Golden section search para minimo em continuo [lo, hi]
398 template<typename F>
938 double golden_min(double lo, double hi, F f, int iters = 200) {
84E const double phi = (sqrt(5.0) - 1.0) / 2.0; // ≈ 0.618
0DD double m1 = hi - phi * (hi - lo);
86C double m2 = lo + phi * (hi - lo);
87F double f1 = f(m1), f2 = f(m2);
586 for (int i = 0; i < iters; i++) {
F4D if (f1 < f2) {
D90 hi = m2; m2 = m1; f2 = f1;
```

```
251     m1 = hi - phi * (hi - lo); f1 = f(m1);
735 } else {
B6E     lo = m1; m1 = m2; f1 = f2;
9FA     m2 = lo + phi * (hi - lo); f2 = f(m2);
E4A }
DC3 }
E67 return (lo + hi) / 2.0;
2D2 }

// Golden section search para maximo em continuo [lo, hi]
398 template<typename F>
CEA double golden_max(double lo, double hi, F f, int iters = 200) {
8ED     return golden_min(lo, hi, [&](double x) { return -f(x); }, iters);
2E7 }
```

FFT

```
658 using ld = double; // troque por long double para mais precisao
316 using CD = complex<ld>;

/* FFT – Fast Fourier Transform
 * Complexidade: O(N log N)
 *
 * fft(a)      → transforma a[] no lugar (tamanho deve ser potencia de 2)
 * conv(a, b)  → convolucao: c[k] = Σ a[i]*b[k-i]
 *
 * Precisao: seguro se (Σa²+Σb²)*log₂N < 9·10¹⁴.
 * Para coeficientes grandes ou mod, use fft-mod.hpp ou ntt.hpp.
 */
```

```
B4C void fft(vector<CD>& a) {
A5B     int n = a.size(), L = 31 - __builtin_clz(n);
F82     static vector<complex<long double>> R(2, 1);
6B4     static vector<CD> rt(2, 1);
AD8     for (static int k = 2; k < n; k *= 2) {
411         auto x = polar(1.0L, acos(-1.0L) / k);
E92         R.resize(n); rt.resize(n);
1D3         for (int i = k; i < 2*k; i++)
CD4             rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
040     }
808     vector<int> rev(n);
5EB     for (int i = 0; i < n; i++) rev[i] = (rev[i/2] | (i&1) << L) / 2;
EE4     for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i], a[rev[i]]);
657     for (int k = 1; k < n; k *= 2)
1E5         for (int i = 0; i < n; i += 2*k)
0C2             for (int j = 0; j < k; j++) {
CD2                 auto x = (ld*)&rt[j+k], y = (ld*)&a[i+j+k];
219                 CD z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
20A                 a[i+j+k] = a[i+j] - z;
1B0                 a[i+j] += z;
707             }
F60 }

17B vector<ld> conv(const vector<ld>& a, const vector<ld>& b) {
F88     if (a.empty() || b.empty()) return {};
BB8     vector<ld> res(a.size() + b.size() - 1);
E9A     int n = 1 << (32 - __builtin_clz(res.size()));
576     vector<CD> in(n), out(n);
F83     copy(begin(a), end(a), begin(in));
D38     for (int i = 0; i < (int)b.size(); i++) in[i].imag(b[i]);
21A     fft(in);
11C     for (auto& x : in) x *= x;
2FC     for (int i = 0; i < n; i++) out[i] = in[-i&(n-1)] - conj(in[i]);
3D7     fft(out);
74E     for (int i = 0; i < (int)res.size(); i++) res[i] = imag(out[i]) / (4*n);
B50     return res;
B6D }
```

FFT Mod

```
240 #include "fft.hpp"

/* FFT Mod – Convolucao modular com mod arbitrario
 * Complexidade: O(N log N) (~2x mais lento que FFT ou NTT)
 *
 * convMod<mod>(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
 *
 * Entradas devem estar em [0, mod).
 * Seguro se N·log₂N·mod < 8.6·10¹⁴.
 *
 * Alta precisao sem mod: troque cut = 1<15 e remova os % mod.
 */

3F9 template<int mod>
11F vector<long long> convMod(
87D     const vector<long long>& a, const vector<long long>& b) {
F88     if (a.empty() || b.empty()) return {};
CE0     vector<long long> res(a.size() + b.size() - 1);
C69     int B = 32 - __builtin_clz(res.size()), n = 1 << B;
528     int cut = (int)sqrt(mod); // alta precisao: use 1<15 e tire os % mod abaixo
584     vector<CD> l(n), R(n), outs(n), outl(n);
5B0     for (int i = 0; i < (int)a.size(); i++)
8B7         l[i] = CD((int)a[i] / cut, (int)a[i] % cut);
```

```
81B     for (int i = 0; i < (int)b.size(); i++)
F55         R[i] = CD((int)b[i] / cut, (int)b[i] % cut);
628     fft(L); fft(R);
603     for (int i = 0; i < n; i++) {
39D         int j = -i & (n-1);
65E         outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
482         outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / CD(0, 1);
1C6     }
41B     fft(outl); fft(outs);
6D1     for (int i = 0; i < (int)res.size(); i++) {
// alta precisao: tire os % mod nas linhas abaixo
852         long long av = (long long)(real(outl[i]) + .5) % mod;
8CD         long long bv = (long long)(imag(outl[i]) + .5)
792             + (long long)(real(outs[i]) + .5);
7AE         long long cv = (long long)(imag(outs[i]) + .5);
557         res[i] = ((av * cut + bv) % mod * cut + cv) % mod;
6CB     }
B50     return res;
B73 }
```

NTT

```
/* NTT – Number Theoretic Transform
 * Complexidade: O(N log N)
 *
 * NTT<mod, g>::conv(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
 *
 * Entradas devem estar em [0, mod).
 * mod deve ser primo da forma 2^a·b+1 (garante raiz de unidade de ordem 2^a).
 * g e a raiz primitiva de mod – nao e escolha livre, e fixo para cada mod.
 *
 * Opcoes de {mod, g}:
 * {998244353, 3} → 119·2²³+1, mais comum, N ≤ 2²³
 * {469762049, 3} → 7·2²⁶+1, N ≤ 2²⁶
 * {167772161, 3} → 5·2²⁵+1, N ≤ 2²⁵
 * {2013265921, 31} → 15·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
 * {2281701377, 3} → 17·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
 * {7340033, 5} → 7·2²⁰+1, N ≤ 2²⁰ (menor mod)
 * {9223372036737335297LL, 3} → 549755813881·2²⁴+1, N≤2²⁴ (mod-2⁹³, ll)
 */
```

```
E34 template<ll mod, ll g>
F12 struct NTT {
D3B     static ll fexp(ll a, ll b) {
C8D         ll res = 1; a %= mod;
FF0         for (; b > 0; b >>= 1, a = a*a%mod)
602             if (b & 1) res = res*a%mod;
B50         return res;
B0B     }

936     static void ntt(vector<ll>& a) {
A5B         int n = a.size(), L = 31 - __builtin_clz(n);
D51         static vector<ll> rt(2, 1);
8EE         for (static int k = 2, s = 2; k < n; k *= 2, s++) {
335             rt.resize(n);
277             ll z[] = {1, fexp(g, mod >> s)};
631             for (int i = k; i < 2*k; i++) rt[i] = rt[i/2] * z[i&1] % mod;
D00         }
808     }
5EB     vector<int> rev(n);
EE4     for (int i = 0; i < n; i++) rev[i] = (rev[i/2] | (i&1) << L) / 2;
657     for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i], a[rev[i]]);
1E5     for (int k = 1; k < n; k *= 2)
0C2         for (int i = 0; i < n; i += 2*k)
86E             for (int j = 0; j < k; j++) {
598                 ll z = rt[j+k] * a[i+j+k] % mod, &ai = a[i+j];
4B8                 a[i+j+k] = ai - z + (z > ai ? mod : 0);
D6A                 ai += z - (ai+z >= mod ? mod : 0);
C0C             }
}

46D     static vector<ll> conv(vector<ll> a, vector<ll> b) {
F88         if (a.empty() || b.empty()) return {};
375         int s = a.size()+b.size()-1, n = 1 << (32 - __builtin_clz(s));
A16         a.resize(n); b.resize(n);
875         ntt(a); ntt(b);
DDC         ll inv = fexp(n, mod-2);
F9E         for (int i = 0; i < n; i++) b[-i&(n-1)] = a[i]*b[i]%mod*inv%mod;
C8D         ntt(b);
95A         return {b.begin(), b.begin()+s};
26F     }
CFB }
```

FWHT

```
/* FWHT – Fast Walsh-Hadamard Transform
 * Complexidade: O(N log N)
 *
 * Convolucao bitwise: c[k] = Σ a[i]*b[j] onde i op j = k
 *
 * fwht_conv<'^'>(a, b) → convolucao XOR
 * fwht_conv<'|'>(a, b) → convolucao OR
 * fwht_conv<'&'>(a, b) → convolucao AND
 *
```

```
* Tamanhos sao ajustados para a proxima potencia de 2 automaticamente.
*/

597 template<char op>
823 void fwht(vector<ll>& a, bool inv = false) {
94D     int n = a.size();
980     for (int len = 1; len < n; len *= 2)
EBC         for (int i = 0; i < n; i += 2*len)
7AB             for (int j = 0; j < len; j++) {
032                 ll u = a[i+j], v = a[i+j+len];
FBD                 if (op == '^') a[i+j] = u+v, a[i+j+len] = u-v;
02F                 if (op == '|') a[i+j+len] = v + (inv ? -u : +u);
729                 if (op == '&') a[i+j] = u + (inv ? -v : +v);
1C4             }
163         if (op == '^' && inv) for (auto& x : a) x /= n;
643 }

597 template<char op>
E0F vector<ll> fwht_conv(vector<ll> a, vector<ll> b) {
43E     int n = 1;
3B9     while (n < (int)max(a.size(), b.size())) n *= 2;
A16     a.resize(n); b.resize(n);
357     fwht<op>(a); fwht<op>(b);
34E     for (int i = 0; i < n; i++) a[i] *= b[i];
9D7     fwht<op>(a, true);
3F5     return a;
22D }
```

String

KMP

```
/* KMP (Knuth-Morris-Pratt)
 * failure[i] = tamanho do maior prefixo proprio de s[0..i] que e tambem sufixo.
 * Complexidade: O(N) build, O(N + M) busca.
 *
 * Aplicacoes:
 * - Busca de padrao P em texto T em O(|P| + |T|).
 * - Menor periodo da string: N - failure[N-1].
 * - Contar ocorrencias de todos os prefixos como sufixos.
 */

090 vector<int> kmp_failure(const string& s) {
163     int n = s.size();
D11     vector<int> f(n, 0);
6F5     for (int i = 1; i < n; i++) {
844         int j = f[i - 1];
424         while (j > 0 && s[i] != s[j]) j = f[j - 1];
04B         if (s[i] == s[j]) j++;
199         f[i] = j;
D74     }
ABE     return f;
8B7 }

// Busca todas as ocorrencias de pattern em text. Retorna indices (0-indexed).
751 vector<int> kmp_search(const string& pattern, const string& text) {
365     string s = pattern + "$" + text;
3EC     vector<int> f = kmp_failure(s);
E6D     int p = pattern.size();
02F     vector<int> res;
D6B     for (int i = p + 1; i < (int)s.size(); i++)
43E         if (f[i] == p) res.push_back(i - 2 * p - 1);
B50     return res;
9C4 }
```

Z Function

```
/* Z-Function
 * z[i] = tamanho do maior prefixo de s que coincide com s[i..].
 * z[0] = 0 por convencao.
 * Complexidade: O(N) build.
 *
 * Aplicacoes:
 * - Busca de padrao P em texto T: z-function de P + "$" + T.
 * - Posicoes i onde z[i] == |P| sao ocorrencias de P em T.
 * - Contar ocorrencias de prefixo de tamanho k: contar z[i] >= k.
 * - Menor periodo da string: menor p tal que p divide N e z[p] == N - p.
 */

7AD vector<int> z_function(const string& s) {
163     int n = s.size();
2B1     vector<int> z(n, 0);
A5C     for (int i = 1, l = 0, r = 0; i < n; i++) {
20E         if (i < r) z[i] = min(r - i, z[i - l]);
4ED         while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
B76         if (i + z[i] > r) l = i, r = i + z[i];
3BB     }
070     return z;
197 }

// Busca todas as ocorrencias de pattern em text. Retorna indices (0-indexed).
430 vector<int> z_search(const string& pattern, const string& text) {
```

```

365 string s = pattern + "$" + text;
EA6 vector<int> z = z_function(s);
E6D int p = pattern.size();
02F vector<int> res;
D6B for (int i = p + 1; i < (int)s.size(); i++)
3F4     if (z[i] == p) res.push_back(i - p - 1);
B50 return res;
0FE }

```

String Hash

```

/* String Hashing (Single e Double Hash)
* Hash polinomial para comparacao de substrings em O(1) apos build O(N).
* Complexidade: O(N) build, O(1) query.
* Memoria: O(N)
*
* SingleHash: um par (MOD, BASE) – risco baixo de colisao para uso geral.
* DoubleHash: dois pares independentes – colisao praticamente impossivel.
*
* Uso:
*   SingleHash h(s);
*   h.get(l, r); // hash de s[l..r]
*   h.get(l, r) == h.get(l2, r2); // compara substrings
*/

F3D using ull = unsigned long long;
/* --- Single Hash --- */
55A struct SingleHash {
029     static const ll MOD = (1LL << 61) - 1;
56C     static const ll BASE;
F42     vector<ll> h, pw;
87D     static ll mul(ll a, ll b) {
3A1         return (__uint128_t)a * b % MOD;
30E     }
67F     static ll add(ll a, ll b) {
9F4         a += b;
BF2         return a >= MOD ? a - MOD : a;
4E8     }
82B     SingleHash() {}
38D     SingleHash(const string& s) : h(s.size() + 1, 0), pw(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
4DB             h[i + 1] = add(mul(h[i], BASE), s[i]);
F0F             pw[i + 1] = mul(pw[i], BASE);
0FA         }
231     }
BA8     ll get(int l, int r) const {
128         return add(h[r + 1], MOD - mul(h[l], pw[r - l + 1]));
358     }
4EF };

E2C const ll SingleHash::BASE = {}() {
798     mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
6FB     return uniform_int_distribution<ll>(1e9, SingleHash::MOD - 1)(rng);
7E5 }();

/* --- Double Hash --- */
2AF struct DoubleHash {
172     static const ll MOD1 = (1LL << 61) - 1;
D89     static const ll MOD2 = 1e9 + 9;
2BF     static const ll BASE1, BASE2;
23A     vector<ll> h1, h2, pw1, pw2;
583     static ll mul1(ll a, ll b) { return (__uint128_t)a * b % MOD1; }
577     static ll add1(ll a, ll b) { a += b; return a >= MOD1 ? a - MOD1 : a; }
997     static ll mul2(ll a, ll b) { return (__uint128_t)a * b % MOD2; }
AC6     static ll add2(ll a, ll b) { a += b; return a >= MOD2 ? a - MOD2 : a; }
A3E     DoubleHash() {}
3C9     DoubleHash(const string& s)
01A         : h1(s.size() + 1, 0), h2(s.size() + 1, 0),
4CF         pw1(s.size() + 1, 1), pw2(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
9F0             h1[i + 1] = add1(mul1(h1[i], BASE1), s[i]);
128             h2[i + 1] = add2(mul2(h2[i], BASE2), s[i]);
562             pw1[i + 1] = mul1(pw1[i], BASE1);
961             pw2[i + 1] = mul2(pw2[i], BASE2);
706         }
366     }
// retorna par de hashes de s[l..r] (0-indexed, inclusive)
C89     pair<ll, ll> get(int l, int r) const {
21E         ll v1 = add1(h1[r + 1], MOD1 - mul1(h1[l], pw1[r - l + 1]));
CA9         ll v2 = add2(h2[r + 1], MOD2 - mul2(h2[l], pw2[r - l + 1]));
F8E         return {v1, v2};
F62     }
331 };

BBD const ll DoubleHash::BASE1 = {}() {
798     mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
9D7     return uniform_int_distribution<ll>(1e9, DoubleHash::MOD1 - 1)(rng);
C70 }();

```

```

0BD const ll DoubleHash::BASE2 = {}() {
A93     mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count() + 1);
AB4     return uniform_int_distribution<ll>(1e9, DoubleHash::MOD2 - 1)(rng);
13F }();

```

Trie

```

/* Trie (Prefix Tree)
* Insercao e busca de strings em O(|s|).
* Memoria: O(N * ALPHA) onde N e o total de caracteres inseridos.
* Requisitos: strings sobre o alfabeto [base, base + ALPHA).
*/

71F template<int ALPHA = 26, char BASE = 'a'>
71A struct Trie {
BF2     struct Node {
C02         int ch[ALPHA];
0F9         int cnt;
302         int end;
D7F         Node() : cnt(0), end(0) { fill(ch, ch + ALPHA, -1); }
B69     };
F5B     vector<Node> t;
8E7     Trie() { t.emplace_back(); }
E7D     void insert(const string& s) {
580         int no = 0;
B4F         for (char c : s) {
FE1             int b = c - BASE;
F67             if (t[no].ch[b] == -1) {
06C                 t[no].ch[b] = t.size();
83D                 t.emplace_back();
6E6             }
C8F             no = t[no].ch[b];
429             t[no].cnt++;
098         }
0B3         t[no].end++;
1A2     }
BA9     bool search(const string& s) const {
580         int no = 0;
B4F         for (char c : s) {
FE1             int b = c - BASE;
1DD             if (t[no].ch[b] == -1) return false;
C8F             no = t[no].ch[b];
1D5         }
A49         return t[no].end > 0;
3BE     }
0B1     int count_prefix(const string& s) const {
580         int no = 0;
B4F         for (char c : s) {
FE1             int b = c - BASE;
FE4             if (t[no].ch[b] == -1) return 0;
C8F             no = t[no].ch[b];
A50         }
395         return t[no].cnt;
EF9     }
4E2     int count(const string& s) const {
580         int no = 0;
B4F         for (char c : s) {
FE1             int b = c - BASE;
FE4             if (t[no].ch[b] == -1) return 0;
C8F             no = t[no].ch[b];
A50         }
1D1         return t[no].end;
34B     }
E20 };

```

Manacher

```

/* Manacher's Algorithm (Palindrome Detection)
* Encontra todos os palindromos em uma string em tempo linear.
* Complexidade: O(N) onde N e o tamanho da string.
* Memoria: O(N)
* Funcoes:
* - manacher(s): arr[i] = tamanho do maior palindromo centrado em i.
* - palindrome: Struct para verificar se s[i..j] e palindromo em O(1).
* - pal_begin(s): arr[i] = tamanho do maior palindromo começando em i.
* - pal_end(s): arr[i] = tamanho do maior palindromo terminando em i.
*/

67A template<typename T>
44D vector<int> manacher(const T& s) {
18F     int l = 0, r = -1, n = s.size();
FC9     vector<int> d1(n), d2(n);
603     for (int i = 0; i < n; i++) {
821         int k = i > r ? 1 : min(d1[l + r - i], r - i);
61A         while (i + k < n && i - k >= 0 && s[i + k] == s[i - k]) k++;
61E         d1[i] = k--;
9F6         if (i + k > r) l = i - k, r = i + k;
950     }
E03     l = 0, r = -1;
603     for (int i = 0; i < n; i++) {

```

```

1A6     int k = i > r ? 0 : min(d2[l + r - i + 1], r - i + 1);
AC1     k++;
A94     while (i + k <= n && i - k >= 0 && s[i + k - 1] == s[i - k])
AC1         k++;
EAA     d2[i] = --k;
26D     if (i + k - 1 > r) l = i - k, r = i + k - 1;
4FE     }
C41     vector<int> ret(2 * n - 1);
E6B     for (int i = 0; i < n; i++) ret[2 * i] = 2 * d1[i] - 1;
E1D     for (int i = 0; i < n - 1; i++) ret[2 * i + 1] = 2 * d2[i + 1];
EDF     return ret;
9CA }

67A template<typename T>
BF0 struct palindrome {
F97     vector<int> man;
B2D     palindrome(const T& s) : man(manacher(s)) {}
9D7     bool query(int i, int j) {
BAD         return man[i + j] >= j - i + 1;
1E7     }
933 };

67A template<typename T>
10D vector<int> pal_begin(const T& s) {
542     int n = s.size() - 1;
E57     vector<int> ret(s.size());
FDE     palindrome<T> p(s);
687     ret[n] = 1;
45B     for (int i = n - 1; i >= 0; i--) {
4AD         ret[i] = min(ret[i + 1] + 2, n - i + 1);
D0D         while (!p.query(i, i + ret[i] - 1)) ret[i]--;
A70     }
EDF     return ret;
4E1 }

67A template<typename T>
934 vector<int> pal_end(const T& s) {
E57     vector<int> ret(s.size());
FDE     palindrome<T> p(s);
D51     ret[0] = 1;
88E     for (int i = 1; i < s.size(); i++) {
A32         ret[i] = min(ret[i - 1] + 2, i + 1);
6EA         while (!p.query(i - ret[i] + 1, i)) ret[i]--;
78E     }
EDF     return ret;
2E4 }

```

Suffix Array (NlogN)

```

/* Suffix Array O(N log N) – Prefix Doubling com Count Sort
* sa[i] = indice do i-esimo sufixo em ordem lexicografica.
* rk[i] = rank do sufixo s[i..] (inverso de sa).
* lcp[i] = LCP entre sa[i-1] e sa[i] (lcp[0] = 0).
* Adiciona '$' ao final da string internamente.
* Complexidade: O(N log N) build, O(|P| log N) search.
*
* Aplicacoes:
* - Busca de padrao P: search(p) retorna {lo, hi} no SA.
* - Substring mais longa repetida: *max_element(lcp).
* - Numero de substrings distintas: N*(N+1)/2 - sum(lcp).
* - LCS de duas strings: SuffixArray::lcs(s, t).
*/

3F4 struct SuffixArray {
AC0     string s;
58B     vector<int> sa, rk, lcp;
264     SuffixArray() {}
357     SuffixArray(string s) : s(s) {
6F2         build();
E8F         build_lcp();
A87     }
0A8     void build() {
FC8         s += '$';
163         int n = s.size();
F27         sa.resize(n); rk.resize(n);
413         vector<pair<char, int>> a(n);
490         for (int i = 0; i < n; i++) a[i] = {s[i], i};
3F8         sort(a.begin(), a.end());
67D         for (int i = 0; i < n; i++) sa[i] = a[i].second;
F27         rk[sa[0]] = 0;
AA4         for (int i = 1; i < n; i++)
C6A             rk[sa[i]] = rk[sa[i - 1]] + (a[i].first != a[i - 1].first ? 1 : 0);
8A4         for (int k = 0; (1 << k) < n; k++) {
8A3             for (int i = 0; i < n; i++) sa[i] = (sa[i] - (1 << k) + n) % n;
188             count_sort();
037             vector<int> new_rk(n);
CCE             new_rk[sa[0]] = 0;
6F5             for (int i = 1; i < n; i++) {
325                 pair<int, int> prv = {rk[sa[i-1]], rk[(sa[i-1] + (1 << k)) % n]};
E09                 pair<int, int> now = {rk[sa[i]], rk[(sa[i] + (1 << k)) % n]};

```

```

1E5         new_rk[sa[i]] = new_rk[sa[i - 1]] + (now != prv ? 1 : 0);
DF8     }
14E     rk = new_rk;
F2D     }
B01 }

00C void count_sort() {
9A8     int n = sa.size();
7C4     vector<int> new_sa(n), cnt(n, 0), pos(n, 0);
CBF     for (auto e : rk) cnt[e]++;
3D7     for (int i = 1; i < n; i++) pos[i] = pos[i - 1] + cnt[i - 1];
1EC     for (auto e : sa) new_sa[pos[rk[e]]++] = e;
253     sa = new_sa;
A28 }

47E void build_lcp() {
9A8     int n = sa.size();
75C     lcp.assign(n, 0);
E74     for (int i = 0, h = 0; i < n - 1; i++) {
0A2         int j = sa[rk[i] - 1];
28E         while (s[i + h] == s[j + h]) h++;
CA3         lcp[rk[i]] = h;
254         if (h) h--;
09C     }
C83 }

// Retorna [lo, hi] no SA onde p ocorre. hi - lo = nº de ocorrencias.
B04 pair<int,int> search(const string& p) const {
B63     int lo, hi, n = sa.size();
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
A3B             if (s.compare(sa[m], p.size(), p) < 0) l = m + 1; else r = m;
103         }
0A8         lo = l;
CB0     }
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
E5A             if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1; else r = m;
D29         }
C1B         hi = l;
277     }
EEF     return {lo, hi};

// LCS entre duas strings. Usa separador que nao aparece em nenhuma delas.
E6C static string lcs(const string& a, const string& b, char sep = '$') {
B29     SuffixArray suf(a + sep + b);
37E     int m = a.size(), best = 0, pos = 0;
564     for (int i = 1; i < (int)suf.sa.size(); i++) {
1D6         bool ga = suf.sa[i] < m, gb = suf.sa[i - 1] < m;
C96         if (ga != gb && suf.lcp[i] > best)
D17             { best = suf.lcp[i]; pos = suf.sa[i]; }
F61     }
D5D     return suf.s.substr(pos, best);
B7C }
48E };

```

Suffix Array (Linear)

```

/* Suffix Array O(N) – SA-IS (Induced Sorting)
* Complexidade: O(N) build. Use apenas se N for muito grande (> 10^6).
* Mais complexo que o O(N log N) – prefira suffix-array-nlogn.hpp.
*
* Campos publicos apos construaao:
* sa[i] = indice do i-esimo sufixo em ordem lexicografica
* rk[i] = rank do sufixo s[i..] (inverso de sa)
* lcp[i] = tamanho do LCP entre sa[i-1] e sa[i]; lcp[0] = 0
*
* Metodos:
* search(p) -> [lo, hi]: intervalo no SA onde p ocorre; hi-lo = nº ocorrencias
*/

FC8 struct SuffixArrayLinear {
AC0     string s;
58B     vector<int> sa, rk, lcp;

CAE     SuffixArrayLinear() {}

B7A     SuffixArrayLinear(const string& s) : s(s + '$') {
C75         vector<int> t(this->s.begin(), this->s.end());
C9A         sa = sa_is(t, 256);
9A8         int n = sa.size();
9BE         rk.resize(n);
66E         for (int i = 0; i < n; i++) rk[sa[i]] = i;
E8F         build_lcp();
B40     }

B04     pair<int,int> search(const string& p) const {
B63         int lo, hi, n = sa.size();
F95         {
993             int l = 0, r = n;
40C             while (l < r) {

```

```

EE4             int m = (l + r) / 2;
A3B             if (s.compare(sa[m], p.size(), p) < 0) l = m + 1; else r = m;
103         }
0A8         lo = l;
CB0     }
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
E5A             if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1; else r = m;
D29         }
C1B         hi = l;
277     }
EEF     return {lo, hi};

void build_lcp() {
9A8     int n = sa.size();
75C     lcp.assign(n, 0);
E74     for (int i = 0, h = 0; i < n - 1; i++) {
0A2         int j = sa[rk[i] - 1];
28E         while (s[i + h] == s[j + h]) h++;
CA3         lcp[rk[i]] = h;
254         if (h) h--;
09C     }
C83 }

template<typename T>
static vector<int> sa_is(const T& s, int sigma) {
163     int n = s.size();
979     if (n == 1) return {0};
979     if (n == 2) return s[0] < s[1] ? vector<int>{0, 1} : vector<int>{1, 0};
A94     vector<bool> t(n);
3A1     t[n - 1] = true;
9C8     for (int i = n - 2; i >= 0; i--)
537         t[i] = s[i] < s[i + 1] || (s[i] == s[i + 1] && t[i + 1]);
0C4     auto is_lms = [&](int i) { return i > 0 && t[i] && !t[i - 1]; };
FC6     vector<int> bkt(sigma + 1, 0);
7E5     for (int c : s) bkt[c + 1]++;
CA7     for (int i = 1; i <= sigma; i++) bkt[i] += bkt[i - 1];
auto induced_sort = [&](const vector<int>& lms) {
386     vector<int> sa(n, -1);
CDA     vector<int> b = bkt;
5EB     for (int i = (int)lms.size() - 1; i >= 0; i--)
F9D         sa[--b[s[lms[i]] + 1]] = lms[i];
332     b = bkt;
ABE     sa[b[s[0]]++] = 0;
69A     for (int i = 0; i < n; i++) if (sa[i] > 0 && !t[sa[i] - 1])
420         sa[b[s[sa[i] - 1]]++] = sa[i] - 1;
332     b = bkt;
452     for (int i = n - 1; i >= 0; i--) if (sa[i] > 0 && t[sa[i] - 1])
3ED         sa[--b[s[sa[i] - 1] + 1]] = sa[i] - 1;
D87     return sa;
1E6 };

vector<int> lms;
for (int i = 0; i < n; i++) if (is_lms(i)) lms.push_back(i);
vector<int> sa = induced_sort(lms);

vector<int> rank_(n, -1);
int cls = 0;
for (int i = 0, prev = -1; i < n; i++) {
FAD     if (!is_lms(sa[i])) continue;
837     if (prev != -1) {
047         bool diff = false;
5B8         for (int d = 0; ; d++) {
F37             if (s[prev+d] != s[sa[i]+d] || t[prev+d] != t[sa[i]+d]) {
28C                 diff = true; break;
305             }
585             if (d > 0 && (is_lms(prev+d) || is_lms(sa[i]+d))) break;
DA4         }
265         if (diff) cls++;
142     }
A6F     rank_[sa[i]] = cls;
C6D     prev = sa[i];
FFD     }

vector<int> lms2, rank2;
for (int i = 0; i < n; i++)
96C     if (rank_[i] != -1) {
9FC         lms2.push_back(i); rank2.push_back(rank_[i]);
839     }

vector<int> sa2 = sa_is(rank2, cls);
vector<int> sorted_lms(lms2.size());
for (int i = 0; i < (int)sa2.size(); i++) sorted_lms[i] = lms2[sa2[i]];
return induced_sort(sorted_lms);
}
}
FAB };

```

Geometry

Point

```

658 using ld = double;
107 const ld eps = 1e-9;
6F1 bool eq(ll a, ll b) { return a == b; }
BFC bool eq(ld a, ld b) { return abs(a - b) <= eps; }

67A template<typename T>
F26 struct Point {
645     T x, y;
0FA     Point(T x = 0, T y = 0) : x(x), y(y) {}

8D2     bool operator<(Point o) const { return tie(x, y) < tie(o.x, o.y); }
511     bool operator==(Point o) const { return eq(x, o.x) && eq(y, o.y); }
9E5     bool operator!=(Point o) const { return !(this == o); }

480     Point operator+(Point o) const { return {x + o.x, y + o.y}; }
229     Point operator-(Point o) const { return {x - o.x, y - o.y}; }
AE8     Point operator*(T k) const { return {x * k, y * k}; }
7C2     Point operator/(T k) const { return {x / k, y / k}; }

61A     T dot(Point o) const { return x * o.x + y * o.y; }
CB8     T cross(Point o) const { return x * o.y - y * o.x; }
// area com sinal do triangulo (this, a, b)
61E     T cross(Point a, Point b) const { return (a - *this).cross(b - *this); }
F68     T dist2() const { return x * x + y * y; }

AD2     ld len() const { return hypot((ld)x, (ld)y); }
978     Point<ld> norm() const { ld l = len(); return {x / l, y / l}; }
178     Point perp() const { return {-y, x}; } // rotacao 90° CCW
8CF     Point<ld> rotate(ld a) const { // a em radianos
97E         return {x * cos(a) - y * sin(a), x * sin(a) + y * cos(a)};
CB4     }

539     friend ostream& operator<<(ostream& os, Point p) {
9B9         return os << p.x << " " << p.y;
DB4     }
2F4 };

29C using Pt = Point<ll>;
BA9 using Ptd = Point<ld>;

// angulo entre os vetores p e q
67A template<typename T>
447 ld angle(Point<T> p, Point<T> q) { return atan2((ld)p.cross(q), (ld)p.dot(q)); }

// area com sinal do triangulo (p, q, r) – positivo = CCW
67A template<typename T>
B80 T sarea(Point<T> p, Point<T> q, Point<T> r) { return p.cross(q, r); }

// se p, q, r sao colineares
67A template<typename T>
EB4 bool is_collinear(Point<T> p, Point<T> q, Point<T> r) {
B84     return eq(sarea(p, q, r), (T)0);
0D1 }

Line

C97 #include "point.hpp"
72C struct Line {
2DF     Ptd p, q;
2BD     Line() {}
299     Line(Ptd p, Ptd q) : p(p), q(q) {}
F5B };

// se p pertence ao segmento r
7E1 bool isinseg(Ptd p, Line r) {
AAC     Ptd a = r.p - p, b = r.q - p;
B7E     return eq(a.cross(b), 0) && a.dot(b) < eps;
C3B }

// projecao do ponto p na reta r
15B Ptd proj(Ptd p, Line r) {
BEA     if (r.p == r.q) return r.p;
2FF     Ptd v = r.q - r.p, u = p - r.p;
5C6     return r.p + v * (u.dot(v) / v.dot(v));
2E8 }

// intersecao das retas r e s (retorna {INF,INF} se paralelas)
24B Ptd inter(Line r, Line s) {
CAD     Ptd u = r.q - r.p, v = s.q - s.p, w = s.p - r.p;
018     ld d = u.cross(v);
F2F     if (eq(d, 0)) return {1e18, 1e18};
2C9     return r.p + u * (w.cross(v) / d);
15F }

// se o segmento r intersecta o segmento s
090 bool interseg(Line r, Line s) {
CBC     if (isinseg(r.p, s) || isinseg(r.q, s) ||
826         isinseg(s.p, r) || isinseg(s.q, r)) return true;
D02     auto ccw = [](Ptd a, Ptd b, Ptd c) {
9B9         return (b - a).cross(c - a) > eps;
3ED     };
13C     return ccw(r.p, r.q, s.p) != ccw(r.p, r.q, s.q) &&

```



```

413         ccw(s.p, s.q, r.p) != ccw(s.p, s.q, r.q);
704 }

// distancia do ponto p a reta r
E49 ld disttoline(Ptd p, Line r) {
FF0     return abs((r.q - r.p).cross(p - r.p) / (r.q - r.p).len());
2F5 }

// distancia do ponto p ao segmento r
A20 ld disttoseg(Ptd p, Line r) {
5A8     if ((r.q - r.p).dot(p - r.p) < 0) return (p - r.p).len();
A6E     if ((r.p - r.q).dot(p - r.q) < 0) return (p - r.q).len();
A19     return disttoline(p, r);
9C0 }

// distancia entre dois segmentos
68D ld distseg(Line a, Line b) {
4DF     if (interseg(a, b)) return 0;
3C3     return min({disttoseg(a.p, b), disttoseg(a.q, b),
458         disttoseg(b.p, a), disttoseg(b.q, a)});
A44 }

```

Circle

```

5FC #include "line.hpp"

// centro da circunferencia que passa por a, b, c
880 Ptd getcenter(Ptd a, Ptd b, Ptd c) {
3FB     Ptd m1 = (a + b) / 2, m2 = (a + c) / 2;
1E3     return inter(Line(m1, m1 + (b - a).perp()),
9A7         Line(m2, m2 + (c - a).perp()));
E56 }

// intersecao da circunferencia (c, r) com a reta ab
E6C vector<Ptd> circ_line_inter(Ptd a, Ptd b, Ptd c, ld r) {
B2F     vector<Ptd> ret;
FAD     Ptd u = b - a, v = a - c;
F37     ld A = u.dot(u), B = v.dot(u), C = v.dot(v) - r * r;
1FA     ld D = B * B - A * C;
818     if (D < -eps) return ret;
8B6     ret.push_back(c + v + u * (-B + sqrt(D + eps)) / A);
387     if (D > eps)
359         ret.push_back(c + v + u * (-B - sqrt(D)) / A);
EDF     return ret;
DAE }

// intersecao das circunferencias (a, r) e (b, R)
24A vector<Ptd> circ_inter(Ptd a, Ptd b, ld r, ld R) {
B2F     vector<Ptd> ret;
8C7     ld d = (b - a).len();
1B3     if (d > r + R + eps || d + min(r, R) < max(r, R) - eps) return ret;
398     ld x = (d * d - R * R + r * r) / (2 * d);
A5C     ld y = sqrt(max(0.0, r * r - x * x));
223     Ptd v = (b - a) / d;
CA4     ret.push_back(a + v * x + v.perp() * y);
9DA     if (y > eps)
07B         ret.push_back(a + v * x - v.perp() * y);
EDF     return ret;
0A2 }

```

Polygon

```

5FC #include "line.hpp"

// 2x a area com sinal (>0 = CCW)
67A template<typename T>
CB6 T area2(const vector<Point<T>>& poly) {
D0C     T a = 0;
BC6     int n = poly.size();
830     for (int i = 0; i < n; i++)
F7A         a += poly[i].cross(poly[(i + 1) % n]);
3F5     return a;
1B9 }

// area do poligono
67A template<typename T>
402 ld polarea(const vector<Point<T>>& poly) {
323     return abs((ld)area2(poly)) / 2.0;
3C3 }

// 0 = fora, 1 = dentro, 2 = na borda
// 0(n), funciona para qualquer poligono simples (convexo ou concavo)
67A template<typename T>
EA9 int winding(const vector<Point<T>>& poly, Point<T> p) {
C24     int n = poly.size(), w = 0;
603     for (int i = 0; i < n; i++) {
D79         Point<T> a = poly[i] - p, b = poly[(i+1)%n] - p;
972         if (a.y <= 0 && b.y > 0) {
E31             T c = a.cross(b);
346             if (c > 0) w++;
EAB             else if (c == 0) return 2;
305         } else if (b.y <= 0 && a.y > 0) {
E31             T c = a.cross(b);
6EA             if (c < 0) w--;
EAB             else if (c == 0) return 2;
992         } else if (b.y == 0 && a.y == 0) {

```

```

09D         if (min(a.x, b.x) <= 0 && 0 <= max(a.x, b.x)) return 2;
88E         }
C79     }
2EA     return w != 0 ? 1 : 0;
E31 }

// ponto no interior do casco convexo (CCW, sem colineares), 0(log n)
67A template<typename T>
406 bool inside_hull(const vector<Point<T>>& hull, Point<T> p) {
ACE     int n = hull.size();
D3E     if (n == 1) return p == hull[0];
4AF     if (n == 2) return hull[0].cross(hull[1], p) == 0
14D         && (p - hull[0]).dot(hull[1] - hull[0]) >= 0
E8C         && (p - hull[1]).dot(hull[0] - hull[1]) >= 0;
D16     if (hull[0].cross(hull[1], p) <= 0 ||
F02         hull[0].cross(hull[n-1], p) >= 0) return false;
B9E     int lo = 1, hi = n - 1;
3D1     while (lo + 1 < hi) {
C86         int mid = (lo + hi) / 2;
353         if (hull[0].cross(hull[mid], p) > 0) lo = mid;
8C0         else hi = mid;
575     }
3B9     return hull[lo].cross(hull[lo+1], p) >= 0;
FF3 }

// corta o poligono com a reta r, mantendo pontos p tais que (r.p, r.q, p) e CCW
010 vector<Ptd> cut_polygon(vector<Ptd> poly, Line r) {
B2F     vector<Ptd> ret;
BC6     int n = poly.size();
2B6     auto ccw = [&](Ptd a, Ptd b, Ptd c) { return (b - a).cross(c - a) > eps; };
603     for (int i = 0; i < n; i++) {
C20         if (ccw(r.p, r.q, poly[i])) ret.push_back(poly[i]);
C6F         if (n == 1) continue;
F59         Line s(poly[i], poly[(i + 1) % n]);
DEA         Ptd p = inter(r, s);
A3D         if (isinseg(p, s)) ret.push_back(p);
6D0     }
8A1     ret.erase(unique(ret.begin(), ret.end()), ret.end());
780     if (ret.size() > 1 && ret.back() == ret[0]) ret.pop_back();
EDF     return ret;
FFE }

// se dois poligonos se intersectam - 0(n*m)
B8C bool interpol(const vector<Ptd>& v1, const vector<Ptd>& v2) {
7D1     int n = v1.size(), m = v2.size();
377     for (auto& p : v1) if (winding(v2, p)) return true;
4C6     for (auto& p : v2) if (winding(v1, p)) return true;
830     for (int i = 0; i < n; i++)
A75         for (int j = 0; j < m; j++)
2B6             if (interseg(Line(v1[i], v1[(i+1)%n]),
F1C                 Line(v2[j], v2[(j+1)%m]))) return true;
D1F         return false;
8F8 }

// Pick: I = (area2 - boundary + 2) / 2, boundary = Σgcd(|dx|,|dy|) nas arestas
B9D long long lattice_points(long long area2, long long boundary) {
4EE     return (area2 - boundary + 2) / 2;
32D }

// distancia entre dois poligonos
CCD ld distpol(const vector<Ptd>& v1, const vector<Ptd>& v2) {
F6B     if (interpol(v1, v2)) return 0;
7D1     int n = v1.size(), m = v2.size();
D63     ld ret = 1e18;
830     for (int i = 0; i < n; i++)
A75         for (int j = 0; j < m; j++)
7DB             ret = min(ret, distseg(Line(v1[i], v1[(i+1)%n]),
D87                 Line(v2[j], v2[(j+1)%m])));
EDF     return ret;
B9D }

```

Convex Hull

```

046 #include "polygon.hpp"

// Casco convexo CCW, 0(N log N). Colineares sao descartados.
67A template<typename T>
CA0 vector<Point<T>> convex_hull(vector<Point<T>> pts) {
CA8     int n = pts.size();
3C9     if (n < 2) return pts;
CD7     sort(pts.begin(), pts.end());
9C8     vector<Point<T>> h;
721     h.reserve(2 * n);
107     for (int phase = 0; phase < 2; phase++) {
5C9         int start = h.size();
D9F         for (auto& p : pts) {
C31             while ((int)h.size() >= start + 2) {
531                 auto a = h[h.size()-2], b = h.back();
9F6                 if ((b - a).cross(p - a) > 0) break;
BFB                 h.pop_back();
0ED             }
A49             h.push_back(p);
37A         }
BFB         h.pop_back();

```

```

05C         reverse(pts.begin(), pts.end());
0D0     }
E30     if (h.size() == 2 && h[0] == h[1]) h.pop_back();
81C     return h;
5AD }

// Struct para operacoes 0(log n) em casco convexo (CCW, sem colineares).
// Construir com convex_hull() antes.
6E2 struct ConvexPol {
1C4     vector<Ptd> pol;
C45     ConvexPol() {}
D90     ConvexPol(vector<Ptd> v) : pol(convex_hull(v)) {}

// se o ponto esta dentro do casco - 0(log n)
46F     bool is_inside(Ptd p) {
B1C         int n = pol.size();
03D         if (n == 0) return false;
E08         return inside_hull(pol, p);
66C     }

// ponto extremo: cmp(p, q) = true se p e mais extremo que q
8AA     int extreme(const function<bool(Ptd, Ptd)>& cmp) {
B1C         int n = pol.size();
4A2         auto extr = [&](int i, bool& cur_dir) {
22A             cur_dir = cmp(pol[(i+1)%n], pol[i]);
35F             return !cur_dir && !cmp(pol[(i+n-1)%n], pol[i]);
DDB         };
63D         bool last_dir, cur_dir;
A0D         if (extr(0, last_dir)) return 0;
993         int l = 0, r = n;
EAD         while (l + 1 < r) {
EE4             int m = (l + r) / 2;
F29             if (extr(m, cur_dir)) return m;
44A             bool rel_dir = cmp(pol[m], pol[l]);
72A             if ((!last_dir && cur_dir) ||
D60                 (last_dir == cur_dir && rel_dir == cur_dir)) {
8A6                 l = m;
1F1                 last_dir = cur_dir;
69C             } else r = m;
D8E         }
792         return l;
B61     }

// indice do ponto com maior produto interno com v - 0(log n)
82A     int max_dot(Ptd v) {
E52         return extreme([&](Ptd p, Ptd q) { return p.dot(v) > q.dot(v); });
C6D     }

// tangentes a partir de p (indices {esquerda, direita}) - 0(log n)
D16     pair<int,int> tangents(Ptd p) {
1AF         auto l = [&](Ptd q, Ptd r) {
540             return (r - p).cross(q - p) > eps;
739         };
B2F         auto R = [&](Ptd q, Ptd r) {
F28             return (q - p).cross(r - p) > eps;
2E2         };
FA8         return {extreme(L), extreme(R)};
00E     }
C9B };

```

Closest Pair

```

C97 #include "point.hpp"

/* Closest Pair - Par de pontos mais proximos
* Complexidade: 0(N log N)
*
* closest_pair(pts) -> {i, j} indices do par mais proximo (i < j)
*
* Distancia ao quadrado: (pts[i] - pts[j]).dist2()
* Funciona com coordenadas inteiras (sem precisao flutuante).
*/

67A template<typename T>
CCC     pair<int,int> closest_pair(const vector<Point<T>>& pts) {
CA0         int n = pts.size();
135         vector<int> idx(n);
295         iota(idx.begin(), idx.end(), 0);
658         sort(idx.begin(), idx.end(),
866             [&](int a, int b) { return pts[a].x < pts[b].x; });
3C9         T best = (pts[idx[0]] - pts[idx[1]]).dist2();
D94         pair<int,int> ans = {idx[0], idx[1]};

// set ordenado por (y, indice)
3B1         set<pair<T,int>> active;
637         for (int l = 0, r = 0; r < n; r++) {
FAF             T delta = (T)ceil(sqrt((double)best)) + 1;
E85             while (pts[idx[r]].x - pts[idx[l]].x > delta) {
9CB                 active.erase(pts[idx[l]].y, idx[l]);
63B                 l++;
34B                 delta = (T)ceil(sqrt((double)best)) + 1;
563             }
98F             auto lo = active.lower_bound((pts[idx[r]].y - delta, INT_MIN));
388             auto hi = active.upper_bound((pts[idx[r]].y + delta, INT_MAX));
DC0             for (auto it = lo; it != hi; it++) {

```

```
E0A         T d = (pts[idx[r]] - pts[it->second]).dist2();
C22         if (d < best) { best = d; ans = {idx[r], it->second}; }
A2C     }
0C6     active.insert({pts[idx[r]].y, idx[r]});
F7D     }
591     if (ans.first > ans.second) swap(ans.first, ans.second);
BA7     return ans;
844 }
```

Others

Stress Test

```
P=code #codigo a testar
0=brute #brute correto
C= #checker (vazio=cmp, ou ex: C=checker)
g++ ${P}.cpp -o sol -O2 || exit 1
g++ ${Q}.cpp -o brt -O2 || exit 1
g++ gen.cpp -o gen -O2 || exit 1
[ -n "$C" ] && { g++ ${C}.cpp -o chk -O2 || exit 1; }
for ((i = 1; ; i++)) do
    echo $i
    ./gen $i > in
    ./sol < in > out
    ./brt < in > out2
    [ -n "$C" ] && ./chk in out out2 || cmp -s out out2
    if [ $? -ne 0 ]; then
        echo "-> entrada:"
        cat in
        echo "-> saida code:"
        cat out
        echo "-> saida brute:"
        cat out2
        break
    fi
done
```

Gen

```
42A #define endl '\n'

C8A mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
    // int x = rng();

463 int uniform(int l, int r) {
A7F     uniform_int_distribution<int> uid(l, r);
F54     return uid(rng);
D9E }

E8D int main() {
886     ios::sync_with_stdio(false);
B95     cin.tie(0);

BB3     return 0;
268 }
```

Checker

```
F3B int main(int argc, char* argv[]) {
4DE     ifstream in(argv[1]), out_sol(argv[2]), out_brt(argv[3]);

BB3     return 0; // return 0 for AC, 1 for WA
231 }
```

Hash Function

O hash de cada subseção é calculado com MD5 (3 chars) sobre o código sem comentários e sem espaços em branco. Usado para verificar integridade durante contest.

O hash de uma linha com } é o hash de todo o código desde a { que a abriu.

Para verificar: copie o código, rode o verificador e compare o prefixo.

Hash Verifier

```
DE3 string getHash(string s){
E7F     ofstream("z.cpp") << s;
FC0     system("g++ -E -P -dD -fpreprocessed ./z.cpp"
041         " | tr -d '[:space:]' | md5sum > sh");
F24     ifstream("sh") >> s;
A15     return s.substr(0, 3);
84E }
```

```
E8D int main(){
973     string l, t;
C3E     stack<string> st({""});
C61     while(getline(cin, l)){
54F         t = l;
242         for(auto c : l)
A53             if(c == '{') st.push(""); else
733             if(c == '}') t = st.top()+l, st.pop();
1B9         cout << getHash(t) + " " + l << endl;
DB4         st.top() += t;
C14     }
692 }
```

Reference Tables

Types, Constants & Complexity

type	bits	max	fmt	valor	obs
int	32	≈ 2.1 × 10 ⁹	%d	12! = 479001600	int
uint	32	≈ 4.3 × 10 ⁹	%u	20! ≈ 2.4 × 10 ¹⁸	ll
ll	64	≈ 9.2 × 10 ¹⁸	%lld	⁽³³⁾ ₍₁₆₎ ≈ 1.2 × 10 ⁹	int
ull	64	≈ 1.8 × 10 ¹⁹	%llu	⁽⁶⁶⁾ ₍₃₃₎ ≈ 7.2 × 10 ¹⁸	ll
double	64	10 ³⁰⁸ , ε ≈ 10 ⁻¹⁵	%lf		
long dbl	80	10 ⁴⁹³² , ε ≈ 10 ⁻¹⁸	%Lf		

<i>n</i>	π(<i>n</i>)	<i>n</i>	<i>O</i> viavel
10 ⁴	1229	10 ⁸	<i>O</i> (<i>n</i>)
10 ⁵	9592	10 ⁷	<i>O</i> (<i>n</i> log <i>n</i>)
10 ⁶	78498	5 × 10 ³	<i>O</i> (<i>n</i> ²)
10 ⁷	664579	500	<i>O</i> (<i>n</i> ³)
10 ⁸	5761455	22	<i>O</i> (2 ^{<i>n</i>})
10 ⁹	50847534	11	<i>O</i> (<i>n</i> !)

Combinatória

Somatórios

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = n^2 \frac{(n+1)^2}{4}$$

$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k n^{m+1-k}$$
$$\sum_{i=0}^n c^i = \frac{c^{n+1}-1}{c-1}, \quad \sum_{i=0}^{\infty} c^i = \frac{1}{1-c}, \quad |c| < 1$$
$$\sum_{i=0}^n ic^i = \frac{nc^{n+2}-(n+1)c^{n+1}+c}{(c-1)^2}, \quad \sum_{i=0}^{\infty} ic^i = \frac{c}{(1-c)^2}, \quad |c| < 1$$

$$H_n = \sum_{i=1}^n \frac{1}{i} \approx \ln n + 0.5772, \quad \ln n < H_n < \ln n + 1$$

$$\sum_{i=1}^n H_i = (n+1)H_n - n, \quad \sum_{i=1}^n iH_i = \frac{n(n+1)}{2}H_n - \frac{n(n-1)}{4}$$

Identidades Binomiais

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k} = \frac{n}{k} \binom{n-1}{k-1} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

$$\binom{n}{m} \binom{m}{k} = \binom{n}{k} \binom{n-k}{m-k}, \quad \binom{n}{k} = (-1)^k \binom{k-n-1}{k}$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k, \quad x^n - y^n = (x-y) \sum_{k=0}^{n-1} x^{n-1-k} y^k$$

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

$$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}, \quad \sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$$

$$\sum_{k=0}^n \binom{r+k}{k} = \binom{r+n+1}{n}, \quad \sum_{k=0}^n \binom{n}{k} \binom{k}{m} = \binom{n+1}{m+1}$$

$$\sum_{k=0}^n \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}, \quad \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$

$$\sum_{k=0}^r (-1)^k \binom{n}{k} = (-1)^r \binom{n-1}{r}$$

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k, \quad (1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k$$

Inversao binomial: $f(n) = \sum_k \binom{n}{k} g(k) \Leftrightarrow g(n) = \sum_k (-1)^{n-k} \binom{n}{k} f(k)$

Contagem

Permutacoes com repeticoes $n_1, ..., n_r$: $\frac{n!}{n_1! \cdots n_r!}$

Permutacoes circulares: $(n-1)!$

Stars & Bars: $x_1 + \cdots + x_k = n, \quad x_i \geq 0$: $\binom{n+k-1}{k-1}$

Inclusao-Exclusao: $|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots \pm |A_1 \cap \cdots \cap A_n|$

Sequências Especiais

Catalan: $C_n = \frac{1}{n+1} \binom{2n}{n}$; 1, 1, 2, 5, 14, 42, 132, 429, ...

Conta: arvores binarias com *n* nos, triangulacoes de (*n*+2)-gono, *n* pares de parenteses validos.

Fibonacci: $F_0 = F_1 = 1, \quad F_{-i} = (-1)^{i-1} F_i$

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}$$

$$F_{i+1}F_{i-1} - F_i^2 = (-1)^i \quad (\text{Cassini})$$

$$F_{n+k} = F_k F_{n+1} + F_{k-1} F_n, \quad F_{2n} = F_n F_{n+1} + F_{n-1} F_n$$

$$\gcd(F_m, F_n) = F_{\gcd(m,n)}, \quad \sum_{k=1}^n F_k = F_{n+2} - 1, \quad \sum_{k=1}^n F_k^2 = F_n F_{n+1}$$

$$\begin{pmatrix} F_n & F_{n-1} \\ F_{n-1} & F_{n-2} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n$$

Representacao de Fibonacci: todo *n* tem representacao unica $n = F_{k_1} + \cdots + F_{k_m}, \quad k_i \geq k_{i+1} + 2, \quad k_m \geq 2$.

Stirling 1ª especie $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$: permutacoes de *n* com *k* ciclos.

$$\left[\begin{smallmatrix} n \\ 1 \end{smallmatrix} \right] = (n-1)!, \quad \left[\begin{smallmatrix} n \\ 2 \end{smallmatrix} \right] = (n-1)!H_{n-1}, \quad \left[\begin{smallmatrix} n \\ n \end{smallmatrix} \right] = 1$$

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = (n-1) \left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right] + \left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right], \quad \sum_k \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = n!$$

$$x^{\overline{n}} = \sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k, \quad \left[\begin{smallmatrix} n \\ -1 \end{smallmatrix} \right] = \binom{n}{2}$$

Stirling 2ª especie $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: particoes de *n* em *k* subconjuntos nao vazios.

$$\left\{ \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1, \quad \left\{ \begin{smallmatrix} n \\ 2 \end{smallmatrix} \right\} = 2^{n-1} - 1, \quad \left\{ \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\} = \left[\begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right] = \binom{n}{2}$$

$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = k \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

Bell $B_n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: # particoes de {1,...,*n*} em subconjuntos nao vazios.

$B_0 = 1, B_1 = 1, B_2 = 2, B_3 = 5, B_4 = 15, B_5 = 52, B_6 = 203$

Numeros de Euler $\left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle$: permutacoes de *n* com *k* ascendentes.

$$\left\langle \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\rangle = 1, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n-1 \\ n-k \end{smallmatrix} \right\rangle, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = (k+1) \left\langle \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\rangle + (n-k) \left\langle \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\rangle$$

$$\left\langle \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\rangle = 2^n - n - 1, \quad \left\langle \begin{smallmatrix} n \\ m \end{smallmatrix} \right\rangle = \sum_{k=0}^m (-1)^k \binom{n+1}{k} (m+1-k)^n$$

$$x^n = \sum_{k=0}^n \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle \binom{x+k}{n}$$

Derangements: $D_n = (n-1)(D_{n-1} + D_{n-2})$; $D_n \approx \frac{n!}{e}$

Josephus: $f(1, k) = 0, \quad f(n, k) = (f(n-1, k) + k) \bmod n$

Funções Geradoras

Serie ordinaria: $A(x) = \sum_{i \geq 0} a_i x^i$; **exponencial:** $A(x) = \sum_{i \geq 0} a_i \frac{x^i}{i!}$

$$\frac{1}{1-x} = \sum x^i, \quad e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}$$

$$(1+x)^n = \sum \binom{n}{i} x^i, \quad \frac{x}{1-x-x^2} = \sum F_i x^i$$

$$\frac{1}{2x} (1 - \sqrt{1-4x}) = \sum C_i x^i, \quad \frac{1}{\sqrt{1-4x}} = \sum \binom{2i}{i} x^i$$

Convolucao: $A(x)B(x) = \sum_i \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i$.

Soma acumulada: $B(x) = \frac{1}{1-x} A(x)$ se $b_i = \sum_{j \leq i} a_j$.

Potências Fatoriais & Bernoulli

$$x^{\underline{n}} = x(x-1)\cdots(x-n+1), \quad x^{\overline{n}} = x(x+1)\cdots(x+n-1)$$

$$x^{\underline{n}} = \sum_{k=1}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] (-1)^{n-k} x^k, \quad x^{\overline{n}} = \sum_{k=1}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k$$

$$x^n = \sum_{k=1}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\underline{k}}$$

Numeros de Bernoulli (*B_i* = 0 para *i* impar > 1):

$B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_4 = -\frac{1}{30}, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}$

$$\frac{x}{e^x - 1} = \sum_{i \geq 0} B_i \frac{x^i}{i!}, \quad n! = \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \left(1 + \Theta\left(\frac{1}{n}\right) \right)$$

Teoria dos Números

Euler Phi & Möbius

$\varphi(n)$: quantidade de inteiros em [1,*n*] coprimos com *n*.

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p} \right) = \prod_{i=1}^n p_i^{e_i-1} (p_i - 1)$$

$$\varphi(mn) = \varphi(m)\varphi(n) \quad (\gcd = 1), \quad \sum_{d \mid n} \varphi(d) = n$$

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^r n = p_1 \cdots p_r, & \sum_{d \mid n} \mu(d) = [n = 1] \\ 0 & p^2 \mid n \end{cases}$$

Inversão de Möbius: $g(n) = \sum_{d \mid n} f(d) \Leftrightarrow f(n) = \sum_{d \mid n} \mu\left(\frac{n}{d}\right) g(d)$

$$\frac{1}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\mu(i)}{i^x}, \quad \frac{\zeta(x-1)}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\varphi(i)}{i^x}, \quad \zeta^2(x) = \sum_{i=1}^{\infty} \frac{d(i)}{i^x}$$

GCD, Diofantinas & TCR

$\gcd(a, b) = \gcd(b, a \bmod b)$; $\text{lcm}(a, b) = ab / \gcd(a, b)$

Bezout: $ax + by = \gcd(a, b)$ sempre tem solução inteira. Em geral, $ax + by = c$ tem solução $\Leftrightarrow \gcd(a, b) \mid c$.

TCR: Sistema $x \equiv a_i \pmod{m_i}$ com *m_i* coprimos dois a dois tem solucao unica mod *M* = Π *m_i*:

$$x = \sum_i a_i M_i (M_i^{-1} \bmod m_i), \quad M_i = \frac{M}{m_i}$$

Aritmetica Modular

Inverso modular: $a^{-1} \bmod m$ via $a^{m-2} \bmod m$ (requer *m* primo). Para todos os inversos 1...*n* de uma vez:

$$\text{inv}[1] = 1, \quad \text{inv}[i] = -(m/i) \cdot \text{inv}[m \bmod i] \bmod m$$

Lucas: Para *p* primo, $\binom{n}{k} \bmod p = \binom{n \bmod p}{k \bmod p} \cdot \left(\frac{\lfloor \frac{n}{p} \rfloor}{\lfloor \frac{k}{p} \rfloor} \right) \pmod{p}$

Recorrências & Séries

Teorema Mestre

$T(n) = aT(\frac{n}{b}) + f(n), \quad a \geq 1, \quad b > 1$:

$f(n) = O(n^{\log_b a - \epsilon}) \Rightarrow T(n) = \Theta(n^{\log_b a})$

$f(n) = \Theta(n^{\log_b a}) \Rightarrow T(n) = \Theta(n^{\log_b a} \log n)$

$f(n) = \Omega(n^{\log_b a + \epsilon})$ e $af(\frac{n}{b}) \leq cf(n) \Rightarrow T(n) = \Theta(f(n))$

Séries de Taylor

$$e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}, \quad \ln \frac{1}{1-x} = \sum \frac{x^i}{i}$$

$$\sin x = \sum (-1)^i \frac{x^{2i+1}}{(2i+1)!}, \quad \cos x = \sum (-1)^i \frac{x^{2i}}{(2i)!}$$

$$\arctan x = \sum (-1)^i \frac{x^{2i+1}}{2i+1}, \quad \frac{1}{(1-x)^{n+1}} = \sum \binom{i+n}{i} x^i$$

Teoria dos Grafos

Teoremas

Euler: Circuito Euleriano $\Leftrightarrow$ todos os vertices tem grau par. Caminho Euleriano $\Leftrightarrow$ exatamente 2 vertices de grau impar.

Fórmula de Euler (planar): $V - E + F = 2$. Grafo planar simples: $E \leq 3V - 6$; planar bipartido: $E \leq 2V - 4$. Todo planar tem algum vertice de grau ≤ 5 .

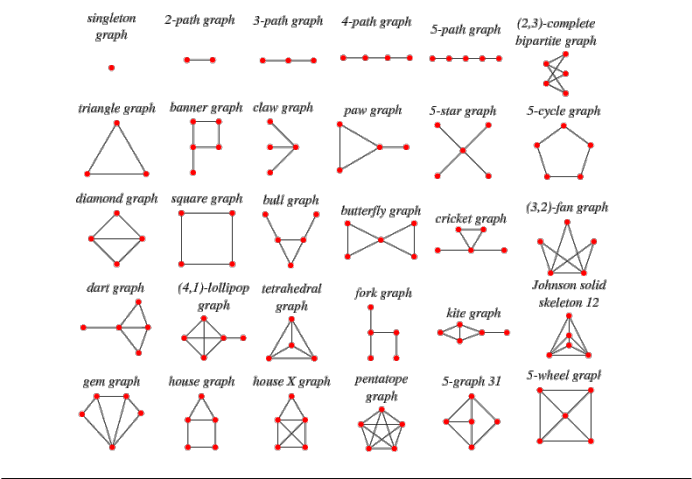
Cayley: Número de árvores geradoras rotuladas de *K_n* e *nⁿ⁻²*.

Kirchhoff: Número de árvores geradoras = qualquer cofator da matriz Laplaciana $L = D - A$.

Konig: Em grafo bipartido: cobertura mínima de vértices = emparelhamento máximo; conjunto independente máximo = V – emparelhamento máximo.
Hall: Existe emparelhamento perfeito de $U \Leftrightarrow \forall S \subseteq U: |N(S)| \geq |S|$.
Menger: Número máximo de caminhos aresta-disjuntos entre s e t = tamanho do corte mínimo s - t .
Dilworth: Em poset, partição mínima em cadeias = anti-cadeia máxima.
Mirsky: Em poset, partição mínima em anti-cadeias = comprimento da maior cadeia.
Prüfer: Bijecao entre árvores rotuladas de n vértices e sequências de comprimento $n-2$ sobre $\{1,...,n\}$.
Erdos-Gallai: Sequência $d_1 \geq ... \geq d_n$ é sequência de graus de algum grafo simples $\Leftrightarrow \sum d_i$ e par e para todo k :

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

Tipos de Grafo



Matemática

Probabilidade

Bayes: Atualiza a probabilidade de A dado que B ocorreu.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Quando $P(B)$ nao e conhecida diretamente (A_j sao hipoteses mutuamente exclusivas e exaustivas):

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Esperanca: $E[X] = \sum xP(X = x)$
Linearidade: $E[aX + bY] = aE[X] + bE[Y]$. Se independentes: $E[XY] = E[X]E[Y]$

Variancia: $\text{Var}(X) = E[X^2] - E[X]^2$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Se independentes: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Markov: $P[X \geq \lambda E[X]] \leq \frac{1}{\lambda}$

Chebyshev: $P[|X - E[X]| \geq \lambda \sigma] \leq \frac{1}{\lambda^2}$

Binomial $B(n, p)$: n ensaios de Bernoulli independentes com prob p .

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad E[X] = np, \quad \text{Var} = np(1 - p)$$

Geometrica $\text{Geom}(p)$: numero de tentativas ate o primeiro sucesso ($q = 1 - p$).

$$P(X = k) = pq^{k-1}, \quad E[X] = \frac{1}{p}, \quad \text{Var} = \frac{q}{p^2}$$

Hipergeometrica $H(N, K, n)$: N itens, K marcados, n sorteados sem reposicao.

$$P(X = k) = \binom{K}{k} \frac{\binom{N-K}{n-k}}{\binom{N}{n}}, \quad E[X] = \frac{nK}{N}$$

Poisson $\text{Pois}(\lambda)$: aproximacao de $B(n, p)$ quando $n \rightarrow \infty, np = \lambda$.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad E[X] = \text{Var}(X) = \lambda$$

Normal $\mathcal{N}(\mu, \sigma^2)$: $p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$, $E[X] = \mu$, $\text{Var} = \sigma^2$.

Coupon collector: n tipos equiprovaveis; numero esperado de sorteios = nH_n .

Paradoxo do aniversario: $\approx 1.18\sqrt{M}$ amostras de $[M]$ para colisao com prob $\geq 50\%$.

Trigonometria

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

$$\sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$e^{ix} = \cos x + i \sin x, \quad e^{i\pi} = -1$$

$$\text{Lei dos senos: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{Lei dos cossenos: } c^2 = a^2 + b^2 - 2ab \cos C$$

Area do triangulo (lados a,b,c; angulos A,B,C):

$$\frac{1}{2} ab \sin C, \quad \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}, \quad \frac{c^2 \sin A \sin B}{2 \sin C}$$

$$\text{Circumraio: } R = ab \frac{c}{4A}, \quad \text{Inraio: } r = \frac{A}{s}$$

Geometria

$$\text{Shoelace: } A = \frac{1}{2} |\sum_i (x_i y_{i+1} - x_{i+1} y_i)|$$

Pick: $A = I + \frac{B^2}{4} - 1$ (I = pontos interiores, B = pontos na borda)

Area por coordenadas: $\frac{1}{2} |\det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}|$

$$\text{Esferas/circulos: } A = \pi r^2, \quad V = \frac{4}{3} \pi r^3$$

Produto vetorial: $\vec{u} \times \vec{v} = (u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x)$

Em 2D: $u_x v_y - u_y v_x$ (> 0 anti-horario, $= 0$ colinear; modulo = area do paralelogramo)

Dist. ponto-ponto: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ (3D: inclui $(z_1 - z_2)^2$)

Dist. ponto-reta ($ax + by + c = 0$): $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Rotacao 2D:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Rotacao 3D em torno de x, y, z por θ ($c = \cos \theta, s = \sin \theta$):

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{pmatrix}, \quad R_y = \begin{pmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{pmatrix}, \quad R_z = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Teoria dos Jogos

Sprague-Grundy: Todo jogo imparcial equivale a um Nim de um unico monte. O valor de Grundy (nimber) de uma posicao u e:

$$g(u) = \text{mex}\{g(v) : v \text{ alcancável de } u\}$$

onde $\text{mex}(S)$ e o menor inteiro nao negativo fora de S .

Posição perdedora $\Leftrightarrow g(u) = 0$. Em jogo composto de partes independentes:

$$g = g_1 \oplus g_2 \oplus \dots \oplus g_k$$

Nim: k montes de tamanhos $a_1, ..., a_k$. Posição perdedora $\Leftrightarrow a_1 \oplus \dots \oplus a_k = 0$.

Nim misere (quem pegar o ultimo *perde*): subtraia 1 de cada pilha; se o XOR resultante $\neq 0$, a posicao original e ganhadora.

Wythoff: 2 montes (a, b) , $a \leq b$. Movimentos: remover qualquer qtd de um monte, ou a mesma qtd dos dois. Posição perdedora $\Leftrightarrow a = \lfloor \varphi k \rfloor$, $b = \lfloor \varphi^2 k \rfloor$ para algum $k \geq 0$, onde $\varphi = \frac{1+\sqrt{5}}{2}$.

Tabelas Numéricas

Números Primos

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73
79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181
191 193 197 199 211 223 227 229 233 239 241 251 257 263 269 271 277 281 283 293 307
311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421 431 433
439 443 449 457 461 463 467 479 487 491 499 503 509 521 523 541 547 557 563 569 571
577 587 593 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 691 701
709 719 727 733 739 743 751 757 761 769 773 787 797 809 811 821 823 827 829 839 853
857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991 997

Fibonacci

n	F_n	n	F_n	n	F_n	n	F_n	n	F_n	n	F_n	n	F_n
0	0	1	1	2	1	3	2	4	3	5	5	6	8
7	13	8	21	9	34	10	55	11	89	12	144	13	233
14	377	15	610	16	987	17	1597	18	2584	19	4181	20	6765

Triângulo de Pascal $\binom{n}{k}$

$n \setminus k$	0	1	2	3	4	5	6	7	8	9	10
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

Debugging

```
-g++ -O2 -std=c++17 -Wall -Wextra  
-fsanitize=address,undefined  
-fno-sanitize-recover=all -g
```

Antes de submeter:

- Escreva casos simples se o sample não for suficiente.
- Limite de tempo apertado? Gere casos máximos.
- Memória está ok?
- Algo pode dar overflow?
- Está submetendo o arquivo certo?

Wrong Answer:

- Imprima sua solução e outputs de debug.
- Está limpando estruturas entre casos de teste?
- O algoritmo cobre todo o range de entrada?
- Releia o enunciado completo.
- Todos os casos de borda estão corretos?
- Entendeu o problema corretamente?
- Variáveis não inicializadas?
- Overflow?
- Confundindo N/M, i/j, etc.?
- Tem certeza que o algoritmo funciona?
- Que casos especiais você não considerou?
- As funções da STL funcionam como você acha?
- Adicione asserts e reesubmeta.
- Rode em casos criados manualmente.
- Execute o algoritmo num caso simples à mão.
- Explique o algoritmo para um colega.
- Peça para o colega revisar seu código.
- Dê uma volta, vá ao banheiro.
- O formato de saída está correto? (espaços, newlines)
- Reescreva do zero ou peça ao colega.

Runtime Error:

- Testou todos os casos de borda localmente?
- Variáveis não inicializadas?
- Lendo/escrevendo fora do range de algum vetor?
- Algum assert que pode falhar?
- Divisão por zero? (mod 0?)
- Recursão infinita possível?
- Ponteiros ou iteradores invalidados?
- Usando memória demais?

Time Limit Exceeded:

- Tem loop infinito?
- Qual é a complexidade do algoritmo?
- Está copiando dados desnecessariamente? (use referências)
- Tamanho da entrada/saída? (considere scanf)
- Evite map/vector; use arrays/unordered_map.
- O que seus colegas acham do algoritmo?

Memory Limit Exceeded:

- Quanto de memória seu algoritmo realmente precisa?
- Está limpando estruturas entre casos de teste?